

FORACCA 2.0

A hybrid process–statistical framework for predicting where climate-resilient reforestation will hold in Armenia

Technical concept and methodological specification · successor to Output 1.2 (v1, 16 March 2026)

Narek Ohanyan · Acopian Center for the Environment, American University of Armenia · Draft v2.0, 21 September 2026

Abstract. Version 1 of the FORACCA vulnerability model asked a random forest to reproduce the local spread of canopy height from three climate deltas, and then read the answer as a forecast. Re-running its own training table shows why that cannot work: the model scores $R^2 = 0.29$ on a random split and $R^2 = 0.04$ when whole 25 km blocks are held out, while on the same random split a forest given only longitude and latitude scores 0.45. Beyond the training range its output is constant. Version 2 replaces the score with an estimand that can be tested, the probability V that a planted cohort survives and reaches canopy height under a stated climate trajectory, and builds it from five coupled modules: topoclimate and water balance; a two-phase plant-hydraulic failure engine; a panel discrete-time hazard model fitted to observed dieback; an adult-niche, attainable-height, growth and establishment module; and a competing-risk viability integrator feeding a CVaR-robust planting portfolio. Physics decides where observations cannot, observations decide where physics is uncertain, and an area-of-applicability gate arbitrates between them. The validation protocol is fixed before fitting, uncertainty is partitioned rather than summarised, and the core formulas below are implemented and unit-tested in the accompanying Python skeleton.

What changes from version 1 to version 2

	Version 1	Version 2
Target	A vulnerability index (FVS) built from climate deltas scaled by the spread of canopy height	Cohort viability V : probability that planted trees survive and reach $H_{\min}$ under trajectory ω
Climate	Epoch-mean deltas, one model, 1 km	Daily forcing, 5+ GCMs $\times$ 4 SSPs, trend-preserving downscaling to 30 m topoclimate; VPD built last from dew-point-downscaled humidity
Physiology	Absent	Two-phase hydraulic failure: stomatal closure, then cuticular loss with a $g_{\min}$ heat phase transition
Observations	One static canopy-height statistic on forest-mask survivors	Annual dieback events and tree-ring growth collapses for every forest pixel present at the start, censored by harvest and fire
Learning	Random 80/20 split, random forest, Gini importances used as coefficients	Spatial-block, forward-chaining, leave-province-out and extrapolation CV; proper scores; monotone GLM/GBM stack; conformal intervals
Extrapolation	Silent: output is flat outside the training range	Explicit: area-of-applicability gate hands over to the mechanistic hazard
Forest ecology	Absent	Establishment hazard, attainable height, growth in physiological age, stand coupling through leaf area, adult niche as consistency test
Uncertainty, decision	One realisation; no decision layer	Factorial ensemble with variance partitioning; CVaR-robust species–site portfolio

1 What version 1 taught us

The v1 note defined a forest vulnerability score, $FVS = \int (\alpha \Delta VPD + \beta \Delta T_{\max} - \gamma \Delta P) \sigma(H)^{-1} dt$, with α, β, γ to be read off random-forest importances, and the repository trained a random forest to predict $\sigma(H)$, the local standard deviation of canopy height, from three growing-season climate deltas on 4 338 one-kilometre pixels. A Streamlit app then added scenario deltas to the table and reported the shortfall of predicted against observed $\sigma(H)$ as vulnerability. It is a working end-to-end prototype, which is exactly why we can be specific about what

Table 1. Audit of the v1 training table and model. All numbers are recomputed from `Armenia_ML_Training_Data.parquet` ($n = 4\,338$ pixels; RF: 500 trees, depth 15, seed 42).

Finding	Evidence	Consequence for v2
Validation leaks space	Random 80/20 split: $R^2 = 0.29$, MAE 1.80 m (predicting the mean: 2.13 m). Blocked 5-fold CV: $R^2 = 0.20, 0.11, 0.04, -0.28, -0.45$ for 5, 10, 25, 50, 75 km blocks. Median nearest-neighbour spacing 0.7 km; neighbours correlate at $r = 0.97\text{--}0.997$ in every feature and 0.58 in the target.	Skill was interpolation between near-duplicates. Block size must come from the data and the CV must mimic the use (§9).
The features are coordinates	The three deltas predict longitude and latitude with $R^2 = 0.87$ and 0.89. Longitude and latitude alone reach $R^2 = 0.45$ on the same random split, and -0.06 at 25 km blocks against 0.04 for the climate features.	A climate model must beat a coordinates-only baseline under blocked CV, or it has learned nothing about climate.
Extrapolation is a constant	Training range of $\Delta T_{\max}$ is 0.05–0.10 K. Adding any warming ≥ 0.25 K changes no prediction (max difference 4×10^{-16} m). Mean FVS is 60.6 % for SSP1-2.6 mid-century and 64.6 % for SSP5-8.5 end-century, against 6.6 % with deltas as observed: the mildest scenario already produces 54 of the 58 points by which the harshest exceeds the no-forcing baseline (Fig. 1b).	Outside the training support the model must hand over to physics, and say so (§7.3).
The label measures structure	$\sigma(H)$ is predicted from mean height alone with $R^2 = 0.67$; its median coefficient of variation is 0.58. At 1 km it mostly reflects cover fraction and stand mosaic, not physiological stress.	The response must be an event that happened to a tree or stand (dieback, growth collapse, planting failure).
Importances are not coefficients	Gini importances are 34.9, 31.4 and 33.7 % for VPD, $T_{\max}$ and P : near-equal, unsigned, and taken from a model that cannot extrapolate. They were described as SHAP values and proposed as α, β, γ .	Effects come from a hazard model with signs and units; ALE and SHAP are used to explain, never to parametrise.
Units and construction	$\Delta VPD / \Delta T_{\max} = 0.83 \text{ kPa K}^{-1}$ in the table; at fixed relative humidity it should be 0.03–0.25. Training VPD (CHELSA) and scenario VPD (CMIP6-derived) are built differently, and the deltas themselves are small (mean $\Delta T_{\max} = 0.07 \text{ K}$).	One VPD construction end to end, checked by a thermodynamic unit test (§5).
No time, no events, survivors only	One row per pixel; no year, no exposure, no mortality event. The forest mask conditions on persistence, so pixels that died are excluded by construction.	Event histories for every pixel that was forest at the start, with harvest and fire as competing events (§7).
Score $\neq$ formula	The app evaluates $\text{clip}((\sigma_{\text{obs}} - \sigma_{\text{pred}}) / \sigma_{\text{obs}} \times 100, 0, 100)$, a model residual, not the time integral of the concept note.	Estimand defined once, in probability units (§2.2).

has to change. We re-ran the delivered table and model (`scripts/audit_v1.py`; the hyper-parameters, seed and split reproduce the published $R^2 = 0.292$ and MAE = 1.80 m exactly), and Table 1 lists what we found.

None of this diminishes what v1 achieved. It settled the data plumbing, the raster conventions, and a demonstrable interface, and it made the central problem legible: when a model is trained on the spatial pattern of a static quantity, it learns geography, and geography does not project. Everything that follows is designed to make that failure impossible to repeat, and to replace a single number with a chain of quantities that can each be checked against something observable.

2 Question, estimand and design principles

2.1 Where the field stands, and what Armenia already tells us

Drought- and heat-induced tree mortality has been documented across forested biomes worldwide (Allen et al., 2010) and is understood well enough mechanistically that it should not be estimated from correlations alone. Syntheses converge on hydraulic failure, often compounded by carbon limitation and biotic attack, as the proximate cause across biomes (Adams et al., 2017; Choat et al., 2018; Hartmann et al., 2018; McDowell et al., 2008), and xylem safety margins are strikingly convergent across forests (Choat et al., 2012). Two recent results sharpen this for our purposes. First, survival time under drought is a two-phase process: stomatal closure delays failure, after which the tree drains its own storage through the cuticle, and in a 16-species temperate experiment the first phase accounted for 85–97 % of total survival time (Gauthey, 2026; Waite et al., 2026), in line with the earlier argument that timely stomatal closure governs resistance (Martin-StPaul et al., 2017). Second, the minimum leaf conductance $g_{\min}$ is not a constant: it rises steeply above a species-specific temperature (Cochard, 2019; Duursma et al., 2019), which is why heat waves can kill trees that drought alone would not. Trait-based hydraulic models such as SurEau (Cochard et al., 2021; Ruffault et al., 2022) embody these ideas, but they need parameters we do not have for Caucasian species, and their probabilities have never been calibrated against Armenian mortality.

The correlative alternative, species distribution models fitted to occurrence, is what most Caucasus studies use (Dağtekin et al., 2020; Pshegusov et al., 2022). It tells us where species are under present climate; it rests on an

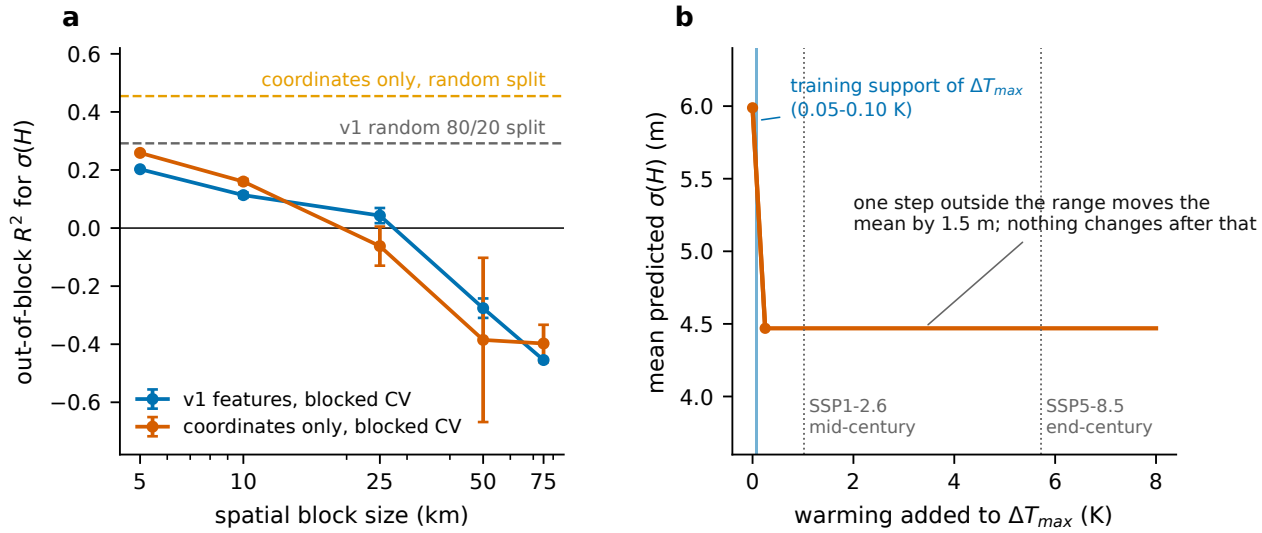


Figure 1. The v1 model, audited on its own data. (a) Out-of-block R^2 against block size for the v1 features (blue) and for coordinates alone (vermillion); dashed lines show the random-split scores. Points are means of three block assignments, bars ± 1 SD. (b) Mean predicted $\sigma(H)$ as extra warming is added to ΔT_{max} with the other two deltas unchanged: one step outside the training range moves the mean by 1.5 m and nothing moves it afterwards.

equilibrium assumption and leaves out demography and dispersal (Guisan and Thuiller, 2005); it is known to fail under non-analogue conditions (Elith et al., 2010; Fitzpatrick and Hargrove, 2009; Pearson and Dawson, 2003), and it says nothing about whether a planted seedling survives its first summers. The productive middle path is hybrid: mechanism supplies the functional form outside the observed range, data supply calibration inside it (Kearney and Porter, 2009; Reichstein et al., 2019).

Local evidence defines what the framework must reproduce. In southern Armenia the mean ring width of *Fagus orientalis* at Shikahogh fell from 1.95 mm in 1965–1993 to 0.89 mm in 1994–2023, with no natural regeneration, while at the northern stand of Teghut it rose from 1.67 to 2.14 mm; local warming runs at 0.026–0.032 °C per year (Stepanyan et al., 2026). Oak and beech in Tavush lose growth and theoretical hydraulic conductivity in extreme dry years and recover within two years, beech faster (Voß et al., 2025); junipers at Khosrov are more drought-sensitive than oak at Tezharuyk (Opáľa-Owczarek et al., 2021). Caucasian distributions are governed by topography and water regime (Pshegusov et al., 2022), and *F. orientalis* is projected to contract sharply with the Caucasus retained as refugium (Dağtekin et al., 2020). These are patterns the calibrated model must pass, not decoration (§6.4). Finally, multi-model work on overshoot pathways shows for the Amazon that the risk of long-term dieback depends on how a warming level is reached and not only on where it ends, beginning at global warming levels as low as 1.3 °C and being largely averted only if temperatures are brought back below 1.5 °C (Munday et al., 2025). A planting decision therefore depends on the trajectory as well as the level, which is why the estimand below is trajectory-explicit.

2.2 The estimand

Let i index a 30 m planting cell, k a planting option (species or provenance, stock type, method), t_0 the planting year and ω one member of the climate ensemble. The quantity we predict is the *cohort viability*

$$V_{ik}(t_0, \omega) = \mathbb{P}[\text{alive at } t_0 + T \text{ and } H \geq H_{\min} \mid \text{climate } \omega] = \left[\prod_{y=1}^T (1 - h_{iky}^{\text{tot}}(\omega)) \right] \cdot \mathbb{P}_{ik}[H_T \geq H_{\min} \mid \text{alive}], \quad (2.1)$$

where h^{tot} is the annual probability that a member of the cohort dies from any cause (§8.2) and the second factor is the probability that a survivor has reached canopy height. Defaults are $T = 30$ yr and $H_{\min} = 10$ m for canopy species, with $T = 15$ and $H_{\min} = 3$ m as a “past browsing and late frost” variant; both are parameters. Every element of Eq. (2.1) is an observable outcome of a planting, so V can be validated against plantation follow-up with a probability sample (§9.5). It is deliberately not a stress index: it is a probability, it has units of “fraction of the cohort”, it compares across species, and a manager can multiply it by hectares.

2.3 Design principles

1. **Estimand before estimator.** Eq. (2.1) is fixed first; every model exists to estimate a factor of it.

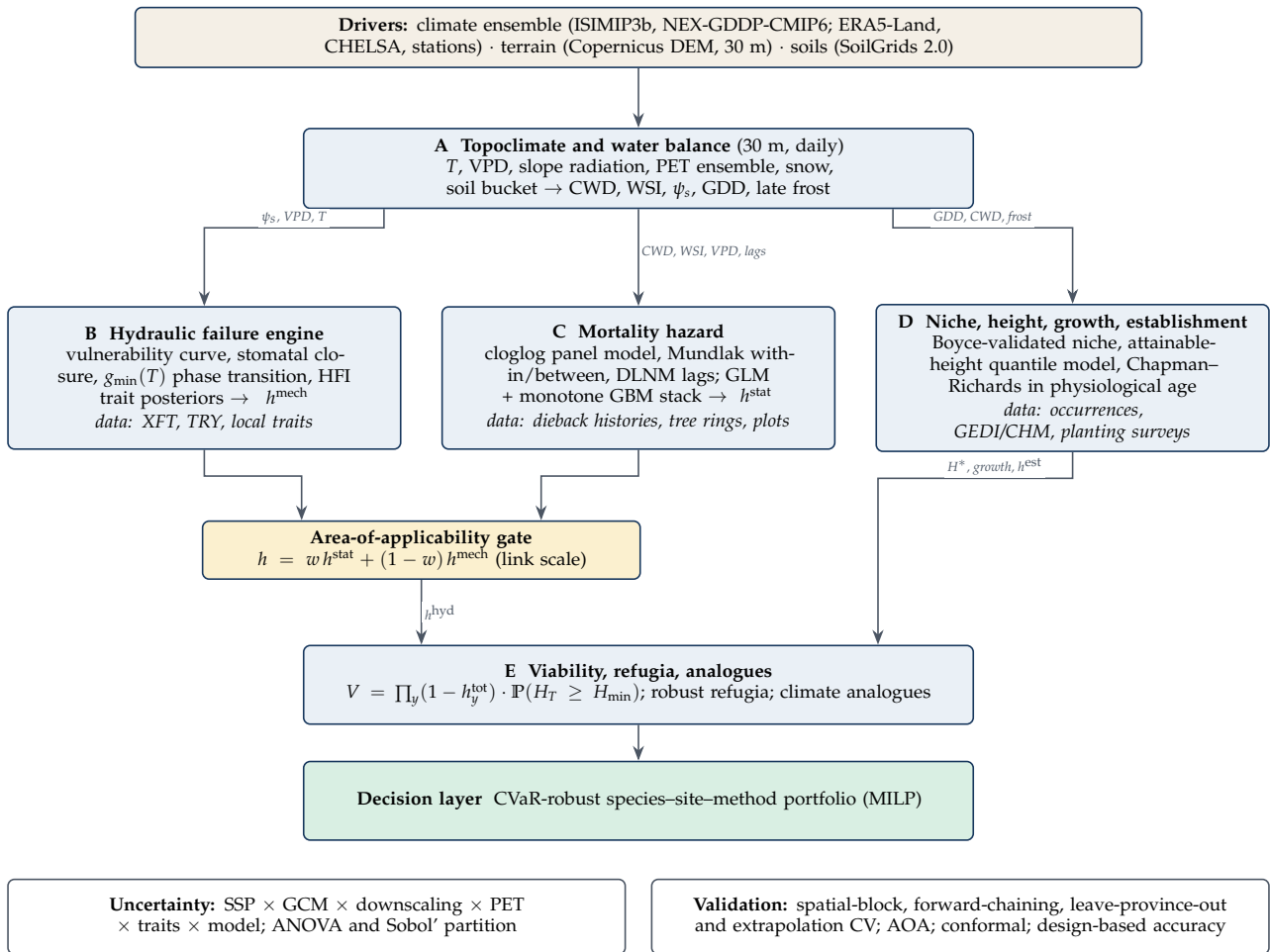


Figure 2. Architecture of FORACCA 2.0. Grey lines carry indicators or hazards between modules; uncertainty propagation and validation apply to every arrow, not only the last one.

2. **Fit hazards to events, not indices to patterns.** The response is something that happened to trees: dieback, growth collapse, planting failure.
3. **Physics where data cannot speak, data where physics is unsure.** The two are blended by an explicit gate, never silently.
4. **Validate the way the model will be used,** in unseen places, unseen years and unseen climates, against baselines that could embarrass it.
5. **Uncertainty is an output.** It is partitioned by source, not summarised as a band.
6. **Decide under uncertainty.** Allocation is optimised for robustness, not for the ensemble mean.
7. **Specify, version, test.** Configuration lives in files, formulas live in tested code, and the validation plan is written down before any model is fitted.

3 Architecture

Figure 2 shows the five modules. **A** converts climate and terrain into daily water and energy balance at 30 m for each ensemble member. **B** turns those indicators into a mechanistic annual failure probability for each species group. **C** fits observed-dieback hazards using the same indicators as covariates and hands over to **B** outside its area of applicability. **D** supplies attainable height, growth in physiological age and the establishment hazard. **E** combines competing hazards into V , derives robust refugia and analogues, and passes ensemble distributions to the decision layer. Uncertainty propagation and validation are cross-cutting, not final steps. Table 2 states which processes each family of approaches represents.

4 Data, harmonisation and the traps we design around

All layers are resampled to one master grid (EPSG:32638, UTM 38N, 30 m). The full ensemble is computed at 100 m; the ensemble median and the decision-scale subsets are recomputed at 30 m. Table 3 lists what each

Table 2. Process coverage of the model families the framework builds on or replaces. “Partial” means represented through proxies or a single assumption.

Process or property	v1	Correlative SDM	Hydraulic model alone	v2
Soil water balance and soil water potential	no	partial	yes	yes
Stomatal-closure timing (phase 1)	no	no	yes	yes
Cuticular loss and heat effect on $g_{\min}$ (phase 2)	no	no	yes	yes
Calibration to observed mortality	no	partial	no	yes
Drought legacies and lags	no	no	no	yes
Establishment bottleneck after planting	no	no	no	yes
Growth to canopy height (time to function)	no	no	no	yes
Defined behaviour outside the training climate	no	no	yes	yes
Uncertainty partitioned by source	no	partial	partial	yes
Robust decision layer	no	no	no	yes

source is for, and what is wrong with it.

Seven harmonisation rules are written into the workflow because each corresponds to a way the same project could go wrong:

1. **One reference period** (1991–2020) for every anomaly, standardised index and climatology, and never a rolling one; a rolling reference standardises the trend away.
2. **Climatologies give pattern, reanalysis and stations give time.** Nothing static is allowed to masquerade as variability.
3. **Humidity travels as vapour pressure.** It is downscaled through the dew point, and VPD is computed last, daily, with the temperature that drives the process in question (§5).
4. **One forest definition for training and projection**, fixed at the start of the record (tree cover in 2000 and the earliest Landsat composite), never by later survival.
5. **Group-aware splits everywhere.** The unit of resampling is the block or the pixel, never the pixel-year.
6. **Every derived quantity is unit-tested against physical expectation**, for example a thermodynamic bound on $\Delta VPD/\Delta T_{\max}$; this would have caught the v1 table.
7. **Every dataset enters through a manifest** (source, version, checksum, licence), and every export script is versioned.

Minimum data for the work plan’s first gate (§11) are: daily forcing 1981–2024; annual dieback labels for $\geq 10^5$ forest pixels over ≥ 10 years; ≥ 200 tree-ring series; trait priors for four functional groups; and a stratified validation sample of ≥ 300 plots (sample-size logic in §9.5).

5 Module A: topoclimate and water balance

Module A produces, for every 30 m cell and every ensemble member, the daily variables that drive everything downstream. Its design goal is that each quantity is computed with the variable that physically controls it, at the resolution at which that variable actually varies.

5.1 Forcing: from GCM to a 30 m day

Bias adjustment. ISIMIP3b fields arrive already trend-preserving (Lange, 2019); residual local bias against the station-corrected ERA5-Land reference (1991–2020) is removed with quantile delta mapping (Cannon et al., 2015). For a future value x_f with model-future, model-historical and observed distributions $F_{m,f}$, $F_{m,h}$, $F_{o,h}$,

$$\begin{aligned}
 \tau &= F_{m,f}(x_f), \\
 \hat{x}_f &= F_{o,h}^{-1}(\tau) \cdot \frac{F_{m,f}^{-1}(\tau)}{F_{m,h}^{-1}(\tau)} \quad (\text{precipitation}), \\
 \hat{x}_f &= F_{o,h}^{-1}(\tau) + [F_{m,f}^{-1}(\tau) - F_{m,h}^{-1}(\tau)] \quad (\text{temperature}),
 \end{aligned} \tag{5.1}$$

applied by month, so that the model’s change in each quantile, and hence in extremes, survives the correction. A plain delta-change factor is retained as a second downscaling branch in the ensemble.

Topoclimate. Daily temperature at cell x is $T(x, d) = T_{\text{ref}}(d) + \Gamma_m [z(x) - z_{\text{ref}}] + \delta_{\text{CAP}}(x, d)$, where the monthly lapse rate Γ_m is regressed from stations (not fixed at -6.5 K km^{-1}) and $\delta_{\text{CAP}} \leq 0$ is a cold-air-pooling term for $T_{\min}$ that scales a terrain concavity index by the share of clear, calm nights. Precipitation follows a monthly

Table 3. Data plan. “Role” states what each layer is used for; “caveat” states what it must not be used for.

Group	Source (resolution)	Role	Caveat
Climate reference	CHELSEA v2.1 (Karger et al., 2017) (1 km); ERA5-Land (Muñoz-Sabater et al., 2021) (0.1°, hourly); Armhydromet stations; TerraClimate (Abatzoglou et al., 2018) (4 km)	Spatial pattern (CHELSA); daily variability and QDM reference (ERA5-Land, stations); independent water-balance benchmark (TerraClimate)	CHELSA is a climatology: never a source of interannual variability. Station network is sparse and needs homogenisation.
Projections	ISIMIP3b (Lange, 2019; Lange and Büchner, 2021): 5 GCMs, SSP1-2.6/3-7.0/5-8.5, 0.5° daily. NEX-GDDP-CMIP6 (Thrasher et al., 2022): 0.25° daily, wide GCM set incl. SSP2-4.5. ScenarioMIP overshoot runs where available	Primary ensemble (ISIMIP3b); SSP2-4.5 and structural breadth (NEX); overshoot stress test of lags and hysteresis	The five ISIMIP3b models span ECS ≈2.6–5.4 K (Zelinka et al., 2020); the top end is a hot model (Hausfather et al., 2022).
Terrain and soils	Copernicus GLO-30 DEM (30 m); SoilGrids 2.0 (Poggio et al., 2021) (250 m, with quantile uncertainty); pedotransfer functions (Saxton and Rawls, 2006)	Slope, aspect, horizon, sky-view, TPI, TWI, cold-air-pooling potential; plant-available water as a random variable	Rooting depth is not observed: treated as a species-specific prior, not a map.
Optical time series	HLS v2 (Claverie et al., 2018) (30 m, 2013–); Landsat archive (30 m, 1985–); kNDVI (Camps-Valls et al., 2021)	Annual canopy-greenness histories → dieback events by temporal segmentation (Kennedy et al., 2010; Zhu and Woodcock, 2014)	Dieback is not defoliation: events must persist ≥ 2 growing seasons.
Structure and disturbance	S2-based height model (v1 layer); global 10 m canopy height (Lang et al., 2023); GEDI footprints (Dubayah et al., 2020); Hansen GFC (Hansen et al., 2013); MODIS burned area	Attainable-height training data; harvest and fire as competing events and masks	Height products saturate and smooth; use for upper envelopes, not stand means.
Ground truth	National Forest Monitoring and Assessment plots; FORACCA plantation surveys (survival and height by planting year and method); tree-ring chronologies (Opáľa-Owczarek et al., 2021; Stepanyan et al., 2026; Voř et al., 2025)	Independent validation sample; establishment hazard; growth–climate functions; extreme-year tests	Design-based sampling must be planned in advance, not opportunistic.
Traits	XFT (Choat et al., 2012), TRY (Kattge et al., 2020), $g_{\min}$ compilation (Duursma et al., 2019); local vulnerability curves, pressure–volume curves and $g_{\min}$ dry-downs	Trait priors and posteriors for four functional groups and ~15 species	Seedling and mature-tree values differ; seedlings need their own class.
Occurrence	GBIF (filtered), plot presences/absences, herbarium records	Adult niche (§8.1)	Presence-only, spatially biased.

elevation gradient with an exposure factor calibrated on CHELSA and the stations. This is where cool, high, shaded refugia first become visible to the model, and where 1 km deltas erase them.

Humidity and VPD. Relative humidity is never interpolated. The reference actual vapour pressure e_a is converted to a dew point, lapsed with elevation, and converted back:

$$e_s(T) = 0.6108 e^{17.27 T / (T + 237.3)}, \quad T_d = \frac{237.3 \ln(e_a / 0.6108)}{17.27 - \ln(e_a / 0.6108)}, \quad (5.2)$$

$$T_d(z) = T_d(z_{\text{ref}}) + \Gamma_d [z - z_{\text{ref}}], \quad e_a(z) = \min\{e_s(T_d(z)), e_s(T_{\min})\},$$

with Γ_d (order -2 K km^{-1}) estimated from station dew points. Three VPDs are then computed, because three processes need three different ones:

$$\begin{aligned} \text{VPD}_{\text{day}} &= e_s(T_{\text{day}}) - e_a, & T_{\text{day}} &= T_{\text{mean}} + 0.45 (T_{\text{max}} - T_{\text{mean}}) \quad (\text{Running et al., 1987}), \\ \text{VPD}_{\text{max}} &= e_s(T_{\text{max}}) - e_a, & \text{VPD}_{24} &= \frac{1}{2} [e_s(T_{\text{max}}) + e_s(T_{\min})] - e_a. \end{aligned} \quad (5.3)$$

Stomatal control uses VPD_{day} ; the upper bound of afternoon demand uses VPD_{max} ; cuticular loss, which continues through the night, uses VPD_{24} . The v1 construction, $e_s(T_{\text{max}})(1 - \text{RH}_{\text{mean}})$, pairs a daily-mean humidity with the daily-maximum temperature and understates afternoon deficit; it is kept in the repository only as a regression test.

5.2 Radiation and potential evapotranspiration

Direct-beam radiation on a slope of inclination β and aspect φ_a scales with the cosine of the incidence angle, $\cos \theta = \cos \beta \cos Z + \sin \beta \sin Z \cos(\varphi_s - \varphi_a)$, with Z and φ_s the solar zenith and azimuth. The slope radiation

is

$$R_{s,\text{slope}} = R_b \frac{\max(\cos \theta, 0)}{\max(\cos Z, \cos 85^\circ)} (1 - s_{\text{horizon}}) + R_d \text{SVF} + R_{\text{refl}}, \quad (5.4)$$

where R_b and R_d split the downscaled global radiation by clearness index, s_{horizon} is the terrain-shading indicator from the horizon angle and SVF is the sky-view factor (Dozier and Frew, 1990). At 40.4°N the clear-sky direct beam on a 25° slope, relative to level ground, is 1.27 for a south and 0.54 for a north aspect at the March equinox (no cast shadows; `slope_radiation_ratio` in the skeleton), a 2.4-fold contrast that is largest in the shoulder seasons, when snowmelt and budburst timing are set.

Potential evapotranspiration is a structural uncertainty, not a number: Penman–Monteith with a fixed surface resistance overstates the atmospheric drying response to warming because it ignores the vegetation and CO₂ response (Milly and Dunne, 2016; Yang et al., 2019). We therefore carry four formulations side by side: FAO-56 reference PM (Allen et al., 1998); Monteith’s form with explicit aerodynamic and canopy conductances (Monteith, 1965),

$$\lambda E = \frac{\Delta (R_n - G) + \rho_a c_p \text{VPD } g_a}{\Delta + \gamma (1 + g_a/g_c)}, \quad g_c = g_{s,\text{max}} f(\text{VPD}) \text{LAI}; \quad (5.5)$$

Priestley–Taylor $\lambda E = \alpha \Delta / (\Delta + \gamma) (R_n - G)$ with $\alpha = 1.26$ (Priestley and Taylor, 1972); and the energy-limited bound $(R_n - G) / \lambda$ (Milly and Dunne, 2016). The CO₂ response enters through $g_{s,\text{max}}$ as a scenario knob (none, moderate, strong).

5.3 Snow, soil water and the stress indicators

Precipitation is partitioned into rain and snow by a logistic function of temperature; snow accumulates and melts by a degree-day rule, and melt and rain form the water input I_d . Soil water is a daily bucket of capacity $W_{\text{max}} = \theta_{\text{avail}} z_{\text{root}}$ (mm), following Granier et al. (1999): with W_d^* the storage after inputs and relative extractable water $\text{REW}_d = W_d^* / W_{\text{max}}$,

$$\begin{aligned} W_d^* &= \min\{W_{d-1} + I_d, W_{\text{max}}\}, & W_d &= W_d^* - \text{AET}_d, \\ \text{AET}_d &= \min\left\{W_d^*, \text{PET}_d \cdot \min(1, \text{REW}_d / \text{REW}_{\text{crit}})\right\}, \end{aligned} \quad (5.6)$$

with $\text{REW}_{\text{crit}} = 0.4$; the excess above W_{max} drains, and the mass balance closes to machine precision (a unit test). From this we derive the annual climatic water deficit and a duration–intensity stress integral,

$$\begin{aligned} \text{CWD} &= \sum_d (\text{PET}_d - \text{AET}_d) \quad (\text{Lutz et al., 2010; Stephenson, 1998}), \\ \text{WSI} &= \sum_d \max\left(0, \frac{\text{REW}_{\text{crit}} - \text{REW}_d}{\text{REW}_{\text{crit}}}\right), \end{aligned} \quad (5.7)$$

and the soil water potential that Module B needs, from the Clapp–Hornberger relation (Clapp and Hornberger, 1978),

$$\psi_s = \psi_{\text{sat}} (\theta / \theta_{\text{sat}})^{-b}, \quad \theta = \theta_{\text{lim}} + (\theta_{\text{fc}} - \theta_{\text{lim}}) W / W_{\text{max}}, \quad (5.8)$$

with texture-dependent ψ_{sat} , b and water contents from pedotransfer functions (Saxton and Rawls, 2006). Plant-available water and rooting depth are sampled (50 draws) from the SoilGrids quantiles and species priors, so soil uncertainty is propagated rather than hidden.

Further indicators are growing degree days above 5 °C, a late-frost index (days with $T_{\text{min}} < -2$ °C after the budburst threshold is reached), heat-spell metrics (days above candidate T_p values; §6) and the multiscalar SPEI (Begu ria et al., 2014; Vicente-Serrano et al., 2010), whose log-logistic distribution is fitted once to 1991–2020 and then applied unchanged to the future.

5.4 Stand coupling

Forest ecology enters Module A through leaf area. Stand density, thinning and species mixture change LAI and rooting depth, hence g_c in Eq. (5.5) and the bucket, and therefore the stress that trees experience. LAI follows the growth module (§8.1); thinning (Sohn et al., 2016) and mixture effects (Grossiord, 2020) are scenario knobs, which makes planting density a genuine decision variable rather than a nuisance parameter.

5.5 Validation of Module A

Downscaled T and P are cross-validated leave-one-station-out; ψ_s and soil moisture are compared with ERA5-Land and satellite soil moisture, CWD with TerraClimate, and all with the unit and closure tests above. Skill in Module A is reported before any ecological model is fitted, so that errors in the physical drivers cannot later be mistaken for ecological signal.

6 Module B: the plant-hydraulic failure engine

Module B is the part of the framework that has to be right where the data run out. It is therefore kept as simple as the physiology allows, fully transparent, and tested against closed-form solutions. It does not simulate a tree; it tracks the one state variable that decides survival in a drought, the water stored between stomatal closure and lethal xylem embolism.

6.1 Vulnerability, closure and the lethal threshold

The xylem vulnerability curve is the logistic form (Pammenter and Van der Willigen, 1998)

$$\text{PLC}(\psi) = \frac{100}{1 + \exp[a(\psi - P_{50})]}, \quad a = \frac{S}{25}, \quad (6.1)$$

where S (% MPa⁻¹) is the slope at P_{50} , since $|d\text{PLC}/d\psi|_{P_{50}} = 25a$. Inverting, the water potential at a given loss L is $\psi_L = P_{50} + a^{-1} \ln[(100 - L)/L]$, so that $P_{88} = P_{50} - a^{-1} \ln(88/12)$. Death is set at $L = 88$ for angiosperms (Urli et al., 2013) and $L = 50$ for conifers (Brodribb and Cochard, 2009), giving the lethal potential $\psi_{\text{crit}} = \psi_L$. Stomata close as the predawn water potential falls, $f_{gs}(\psi) = [1 + \exp\{a_s(\psi - \psi_{gs50})\}]^{-1}$, and we take $\psi_{\text{close}} = \psi_{gs90} = \psi_{gs50} - a_s^{-1} \ln 9$ as the potential at which they are effectively shut.

6.2 Two phases of survival

Phase 1, time to closure, is delivered by Module A: stomata are treated as closed on day d when the root-zone potential satisfies $\psi_{s,d} \leq \psi_{\text{close}}$. Because this phase carries most of the survival time (Gauthey, 2026; Waite et al., 2026), the accuracy of ψ_s , and hence of soil depth, rooting depth and PET, matters more for timing than any trait in phase 2. This is the reason Module A is validated first.

Phase 2, time to critical failure, starts at closure. With stomata shut, the plant loses water only through the cuticle and bark, and draws on its stored water. A mass balance per unit leaf area gives

$$C \frac{d\psi}{dt} = -E_{\min}, \quad E_{\min} = g_{\min}(T) \frac{\text{VPD}_{24}}{P_{\text{atm}}}, \quad B = C(\psi_{\text{close}} - \psi_{\text{crit}}), \quad t_{\text{crit}} = \frac{B}{E_{\min}}, \quad (6.2)$$

with C the effective whole-plant capacitance (mmol m⁻² MPa⁻¹), $g_{\min}$ in mmol m⁻² s⁻¹, $\text{VPD}_{24}/P_{\text{atm}}$ dimensionless (mol mol⁻¹), and B the hydraulic buffer between closure and death (t_{crit} is in seconds, divided by 86 400 for days below). The closed form (cf. Martin-StPaul et al., 2017) is the unit test for the daily engine, which accumulates the index

$$A_d = (1 - \rho \mathbb{1}[\text{open}_d]) A_{d-1} + \mathbb{1}[\text{closed}_d] \cdot 86400 E_{\min,d}, \quad \text{HFI}_d = A_d/B, \quad (6.3)$$

where $\rho \in [0, 1]$ is the fraction of the deficit refilled on a day when stomata are open (rain, or $\psi_s > \psi_{\text{close}}$). Hydraulic failure is defined as $\max_d \text{HFI}_d \geq 1$ within a year.

The heat phase transition. Residual conductance is not constant: it increases slowly with temperature, then abruptly above a critical temperature T_p at which the cuticle loses its barrier properties (Cochard, 2019; Duursma et al., 2019),

$$g_{\min}(T) = \begin{cases} g_{25} Q_a^{(T-25)/10}, & T \leq T_p, \\ g_{25} Q_a^{(T_p-25)/10} Q_b^{(T-T_p)/10}, & T > T_p, \end{cases} \quad (6.4)$$

with $Q_a \approx 1.2$ and $Q_b \approx 4.8$ of the order reported by Cochard (2019), and g_{25} , T_p species-level random variables (T_p spans roughly 30–45 °C across species). A worked example with the placeholder trait set of the skeleton ($C = 3 \times 10^4$, $P_{50} = -3$ MPa, $S = 40$, $\psi_{\text{close}} = -2.5$ MPa, $g_{25} = 3$, $T_p = 38$ °C, angiosperm): $\psi_{\text{crit}} = -4.25$ MPa, $B = 5.2 \times 10^4$ mmol m⁻², and at $\text{VPD}_{24} = 2$ kPa, $P_{\text{atm}} = 90$ kPa, t_{crit} is 8.3 days at 30 °C but 2.8 days at 44 °C. Same tree, same VPD, 14 K hotter, a third of the time; the difference is the phase transition, and it is exactly what a model trained only on the current climate cannot see (Fig. 3b). Trait values in the figure and skeleton are placeholders that exercise the code and are not results.

6.3 From traits to a probability

Trait values differ among individuals and are uncertain to us; the two must not be mixed. We therefore sample in two levels. An outer loop of 50 draws takes hyper-parameters (group means and spreads of P_{50} , S , ψ_{close} , C ,

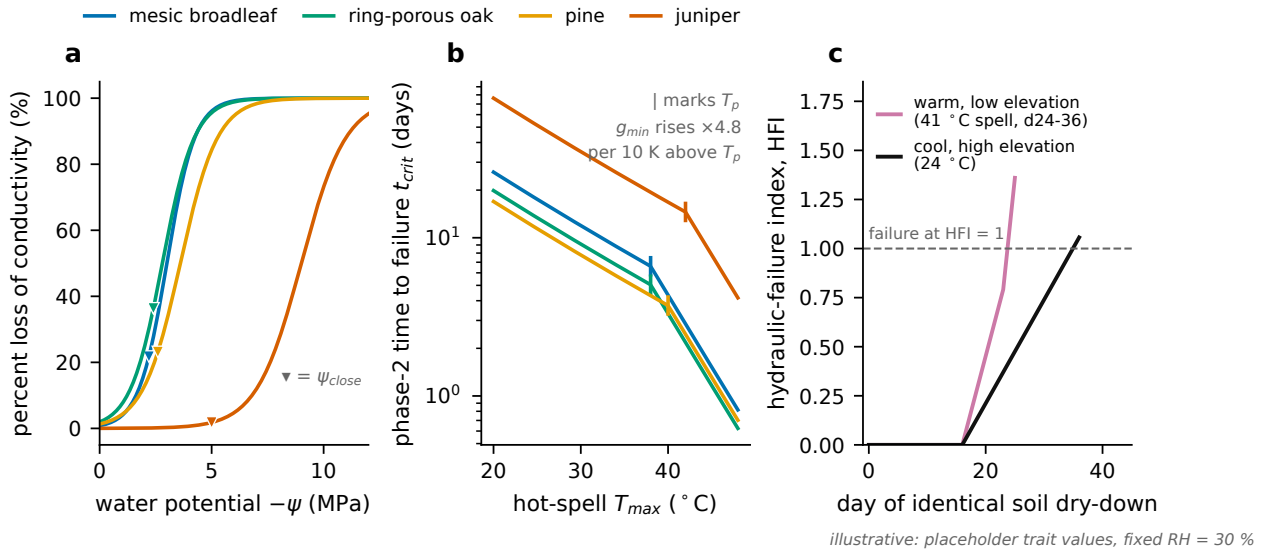


Figure 3. The hydraulic engine, with illustrative placeholder traits for four functional groups. (a) Vulnerability curves; triangles mark the stomatal-closure potential. (b) Phase-2 time to failure against hot-spell temperature at fixed relative humidity; g_{min} steepens the fall above each group's T_p (tick). (c) The same soil dry-down at a warm low site with a 41 °C spell and at a cool high site: closure occurs on the same day, failure much earlier at the warm site.

g_{25} , T_p) from posteriors built on XFT, TRY and the g_{min} compilation and updated with local measurements; an inner loop of 200 draws samples individuals. The mechanistic annual hazard is the share of individuals that fail,

$$h_{iky}^{mech} = \frac{1}{M} \sum_{m=1}^M \mathbb{1} \left[\max_{d \in y} HFI_{iky,d}^{(m)} \geq 1 \right], \quad (6.5)$$

and the spread of h^{mech} across the outer loop is its knowledge uncertainty. Seedlings, established saplings and mature trees form separate trait classes because capacitance per unit leaf area, rooting depth and ψ_{close} change with ontogeny, and because larger trees are more exposed to drought (Bennett et al., 2015).

Emulation. Running the engine for every 30 m cell, year and draw is not feasible, so we emulate it. A Latin hypercube over stress-spell summaries (longest closed spell, mean VPD₂₄ and T during it, number of closed days) and trait-group descriptors is passed through the full engine, and a gradient-boosting regressor with monotone constraints (increasing in spell length, VPD and temperature; decreasing in C and B) is fitted to h^{mech} . Monotonicity is guaranteed by construction, and the emulator must reproduce the full engine on a held-out hypercube to within 0.02 in absolute hazard before it is used.

6.4 Pattern-oriented calibration, and what Module B does not claim

Prior trait draws are constrained by the patterns that Armenia already exhibits, using approximate Bayesian computation (Beaumont, 2010; Grimm et al., 2005). A draw θ is run over 2000–2024 forcing at reference sites and accepted if its summary statistics fall within a tolerance of the observed ones: (i) the Boyce index of predicted viability against present-day adult occurrence for each group; (ii) rank correlation between engine-predicted failure years and tree-ring growth collapses at the chronology sites (Opáľa-Owczarek et al., 2021; Stepanyan et al., 2026; Voß et al., 2025); (iii) observed dieback frequency by elevation band; and (iv) published orderings from dry-down experiments (Waite et al., 2026). The posterior narrows the priors; whatever mismatch remains is absorbed by a two-parameter recalibration on the link scale, $\text{cloglog}(\tilde{h}^{mech}) = a_g + b_g \text{cloglog}(h^{mech})$ with $b_g > 0$, fitted per functional group on the observed event panel with hierarchical shrinkage. A monotone map of the link cannot change the shape of the physical response, so extrapolation keeps its mechanistic form.

The engine assumes constant capacitance, a plant hydraulically isolated from the soil once stomata close, no explicit carbon-starvation pathway, leaf temperature equal to T_{max} (tested against a leaf–air offset), and a single hydraulic threshold per group. Each is a documented limitation and a sensitivity experiment; none is hidden in the code.

7 Module C: a hazard model fitted to observed mortality

Module B says what should happen to a tree of a given type. Module C says what has happened to the forests of Armenia, and it is the only part of the framework that is calibrated against outcomes. It is a discrete-time

survival model on a panel of pixel-years.

7.1 Response and population at risk

The population at risk is every 30 m pixel that was forest at the start of the record, followed until a dieback event, until censoring by harvest or fire (which are competing events, handled in §8.2), or until the end of 2024. This removes the survivorship bias of v1: pixels that died stay in the data as events. A dieback event in year t is a standardised anomaly of annual maximum kNDVI, relative to the pixel's trailing 10-year median, below -2 that has not recovered above -1 within two seasons and coincides with no disturbance flag; the segmentation algorithm (LandTrendr or CCDC; Kennedy et al. 2010; Zhu and Woodcock 2014) and thresholds are fixed in the configuration and versioned. Because satellite dieback is an imperfect measure of mortality, sensitivity Se and specificity Sp are estimated on a stratified sample of interpreted pixel-years, and the observed hazard is corrected with the standard misclassification identity (Rogan and Gladen, 1978),

$$h_{\text{obs}} = Se h + (1 - Sp)(1 - h) \implies h = \frac{h_{\text{obs}} - (1 - Sp)}{Se + Sp - 1}, \quad (7.1)$$

or the same relation is embedded in the likelihood. Tree-ring growth collapses (ring width below 40 % of the trailing 10-year median) form a second, independent response on the chronology network; they are used to test temporal skill and to link the remote-sensing definition to physiological stress.

7.2 The model

Let $h_{i,t} = \mathbb{P}(T_i = t \mid T_i \geq t, \text{history})$ be the annual hazard of pixel i . We use the complementary log-log link, the discrete-time counterpart of proportional hazards (Prentice and Gloeckler, 1978; Singer and Willett, 2003):

$$\text{cloglog}(h_{i,t}) = \ln[-\ln(1 - h_{i,t})] = \ln \Delta_{i,t} + \eta_{i,t}, \quad S_i(t) = \prod_{s \leq t} (1 - h_{i,s}) = \exp\left(-\sum_{s \leq t} \Delta_{i,s} e^{\eta_{i,s}}\right), \quad (7.2)$$

where the offset $\ln \Delta$ lets periods of unequal length (censoring within a year) enter one likelihood, and the identity on the right is exact. The linear predictor is

$$\eta_{i,t} = \beta_0 + f(a_{i,t}) + \beta_W^\top \tilde{x}_{i,t} + \beta_B^\top \tilde{x}_i + \sum_{j=1}^J \sum_{k=1}^K \theta_{jk} \sum_{\ell=0}^L b_j(z_{i,t-\ell}) c_k(\ell) + \gamma^\top \mathbf{u}_{i,t}. \quad (7.3)$$

Its parts have distinct jobs.

Within and between (Mundlak). Each climate covariate is split into a place mean and a departure, $\tilde{x}_i = T_i^{-1} \sum_t x_{i,t}$ and $\tilde{x}_{i,t} = x_{i,t} - \tilde{x}_i$ (Bell and Jones, 2015; Mundlak, 1978). β_W is the response to year-to-year departures at a fixed place, identified from within-place variation and therefore robust to any unmeasured, time-invariant site quality; it is the closest thing in the data to the effect of weather on mortality. β_B describes differences between places and contains adaptation, stand structure and everything else that differs across sites. Confounding of the two is how space-for-time substitution goes wrong, so the model keeps them apart.

Drought legacies. The cross-basis term is a distributed-lag non-linear model (Gasparrini et al., 2010): the exposure z (water-stress integral or CWD anomaly) enters through a spline basis b_j and its effect is spread over lags $\ell = 0, \dots, L$ ($L = 4$ yr) through a second basis c_k . Drought legacies of several years are pervasive in forests (Anderegg et al., 2015), and a model without lags mistakes them for noise.

Stand terms. $a_{i,t}$ is time since disturbance or planting, and $\mathbf{u}_{i,t}$ contains canopy height class, species-group composition and stand density proxies, with a size $\times$ stress interaction whose sign is constrained a priori (Bennett et al., 2015). Coordinates and raw elevation are excluded: anything geographic must reach the model through a physical driver, or it is a place label that cannot be projected.

Rare events. All events are kept; non-events are sampled within blocks with known fraction π_0 and weighted by $1/\pi_0$, which gives a consistent estimator of the full-data model (King and Zeng, 2001). Uncertainty comes from a block bootstrap, not from the pixel-level standard errors, which are meaningless under spatial dependence.

7.3 Learners, stacking and the gate

Three learners share the panel: (i) an elastic-net cloglog GLM, transparent and extrapolating linearly on the link scale, which is at least an assumption one can read; (ii) gradient boosting with monotone constraints on physically signed covariates (hazard non-decreasing in CWD, VPD, heat and stress duration; non-increasing in plant-available water); and (iii) as a benchmark only, a random forest, which cannot extrapolate. Out-of-fold link-scale predictions of (i) and (ii) are combined by non-negative least squares with weights summing to one

(Breiman, 1996; Wolpert, 1992), inside a nested cross-validation so that stacking weights are never fitted on the data they are scored on. The result, h^{stat} , is trustworthy only where the data are.

To decide where, we compute the dissimilarity index of Meyer and Pebesma (2021): the distance from a prediction cell to its nearest training cell in a standardised, importance-weighted covariate space, divided by the mean pairwise training distance, $\text{DI}(x) = \min_j \|x - x_j\| / \bar{d}$. The threshold DI^* is the upper whisker of the training cells' own DI values, each computed against training cells in *other* cross-validation folds, so that it reflects the distances over which the model was actually validated. The final hazard blends on the link scale,

$$\begin{aligned} \text{cloglog}(h) &= w \text{cloglog}(h^{\text{stat}}) + (1 - w) \text{cloglog}(\tilde{h}^{\text{mech}}), \\ w &= \exp[-\kappa \max\{0, \text{DI}/\text{DI}^* - 1\}], \quad \kappa = 2, \end{aligned} \quad (7.4)$$

so that $w = 1$ inside the area of applicability, decays smoothly outside it, and cumulative hazards, not probabilities, are what get mixed. The share of cells and years outside the area of applicability is reported as an output for every scenario, the *extrapolation footprint*: a statement of how much of the projected future rests on physics rather than on observation.

7.4 Bounding the space-for-time assumption

Projecting the within and between effects requires a decision about whether trees adapt to the shifted mean. We do not make it; we bracket it. With $x_{i,t}^{\text{proj}}$ the projected covariate and $\bar{x}_i^{\text{hist}}$, $\bar{x}_i^{\text{proj}}$ the historical and projected 20-year place means,

$$\eta_{i,t}^{\text{no adapt}} : \tilde{x} = x_{i,t}^{\text{proj}} - \bar{x}_i^{\text{hist}}, \quad \bar{x} = \bar{x}_i^{\text{hist}}; \quad \eta_{i,t}^{\text{full acclim}} : \tilde{x} = x_{i,t}^{\text{proj}} - \bar{x}_i^{\text{proj}}, \quad \bar{x} = \bar{x}_i^{\text{proj}}. \quad (7.5)$$

In the first, the within-place response applies to the whole climatic shift; in the second, β_B , typically weaker because trees are adapted to their local mean, applies to the shift of the mean. Every projection is reported as this bracket, and the width between the bounds is itself a result: it measures how much of the future depends on an ecological assumption that no dataset can currently settle.

8 Modules D and E: forest ecology, growth and cohort viability

8.1 Module D: niche, attainable height, growth and establishment

Adult niche as a consistency test. For each species we fit a penalised presence–background model on physically meaningful predictors (CWD, growing degree days, winter minimum, VPD, soil), with the background drawn from a target-group sample to absorb collection bias, and score it under spatial-block CV with the continuous Boyce index (Hirzel et al., 2006). With $P/E_b = (n_b^{\text{pres}}/n^{\text{pres}})/(n_b^{\text{bg}}/n^{\text{bg}})$ in suitability bin b , $\text{CBI} = \rho_s(b, P/E_b)$, the Spearman rank correlation across bins, which needs no presence–absence data and no threshold. The niche is deliberately not a driver of V . It is a realised niche, shaped by competition, dispersal and land use; in the Caucasus competition alone contracts fundamental niche areas by a factor of 1.2–1.7 (Pshegusov et al., 2022), so it would veto exactly the sites where planting is most useful. Its job is consistency: sites with high V should be enriched among adult occurrences, and the mismatch map separates climatic limitation from dispersal or land-use limitation.

Attainable height. Observed canopy height reflects age and disturbance history as much as site quality, so its mean is a poor measure of potential. We estimate the upper envelope, as in site-index work (Skovsgaard and Vanclay, 2008), by quantile regression (Cade and Noon, 2003) of GEDI and canopy-height-model heights on climate, soil and terrain,

$$H^*(x) = \hat{f}_\tau(x), \quad \hat{f}_\tau = \arg \min_f \sum_i \rho_\tau(H_i - f(x_i)), \quad \rho_\tau(u) = u(\tau - \mathbb{1}[u < 0]), \quad \tau = 0.9, \quad (8.1)$$

scored by pinball loss and by coverage of the τ -quantile on held-out blocks, with conformalised quantile regression intervals (Romano et al., 2019). Outside the area of applicability, H^* falls back on a water-limited scaling by the transpiration ratio AET/PET from Module A.

Growth in physiological age. Height follows the Chapman–Richards curve in a physiological age that advances more slowly in stress years,

$$H(a) = H^* (1 - e^{-ka})^p, \quad a_{y+1} = a_y + \varphi_y, \quad \varphi_y = f_W f_T f_F, \quad f_W = \frac{1}{1 + (\text{CWD}_y / \text{CWD}_{50})^m}, \quad (8.2)$$

where f_T is a growing-degree-day modifier, f_F a late-frost modifier, $\varphi_y \in (0, 1]$ (Liebig-type limitation as in 3-PG; Landsberg and Waring 1997) and CWD_{50} , m are estimated from the tree-ring growth–CWD relation. Stress years delay development without reversing it, which keeps the closed form. The age at which a cohort reaches $H_{\min}$ is $a^* = -k^{-1} \ln[1 - (H_{\min}/H^*)^{1/p}]$ (infinite if $H_{\min} \geq H^*$), and $\mathbb{P}[H_T \geq H_{\min}] = \mathbb{P}[\sum_{y \leq T} \varphi_y \geq a^*]$ is evaluated over ensemble members and parameter draws. This is also where Module A gets its leaf area: LAI follows $H(a)$, closing the loop between stand development and stand water use.

Establishment. The regeneration niche is narrower than the adult niche (Davis et al., 2019; Dobrowski et al., 2015; Jackson et al., 2009), and the pattern is visible locally: the southern beech stand shows mature trees and no regeneration (Stepanyan et al., 2026). Planted seedlings pass through the same bottleneck, with shallow roots and little storage. Establishment is therefore a discrete-time hazard over the years until height exceeds a vulnerability height H_{est} ,

$$h_{\tau}^{\text{est}} = \text{cloglog}^{-1}(\beta_0 + \beta^{\top} x_{\tau} + g(\tau)), \quad S_{\text{est}} = \prod_{\tau} (1 - h_{\tau}^{\text{est}}), \quad (8.3)$$

with covariates in the first seasons after planting: minimum topsoil REW, peak VPD_{day}, heat days, late frosts, snow-cover duration, stock type and size, browsing protection and planting method. Data are plantation follow-up surveys and regeneration plots. The window length $\tau \leq \tau_{\max}$ is set by the growth module, so that a site that slows growth also prolongs exposure.

8.2 Module E: cohort viability, refugia and analogues

Competing hazards. The all-cause annual hazard combines the mechanism-specific hazards $h_{k,y}$ for establishment, hydraulic failure, late frost, fire and biotic agents,

$$h_y^{\text{tot}} = 1 - \prod_k (1 - h_{k,y}), \quad V = \left[\prod_{y=1}^T (1 - h_y^{\text{tot}}) \right] \cdot \mathbb{P}[H_T \geq H_{\min}], \quad (8.4)$$

which is Eq. (2.1) with its factors named. h_{hyd} is the gated hazard of Eq. (7.4); h_{frost} and h_{fire} are cloglog models on the late-frost index and on a fire-weather index with burned-area records; h_{bio} (insects, disease, browsing) is a baseline from long-term plot mortality and is not projected with climate, which we state as a limitation rather than pretend otherwise. The product assumes independence conditional on covariates; it is tested by comparing the modelled all-cause hazard with observed all-cause plantation mortality, and compound events such as drought followed by fire enter as interactions in Module C.

Robust refugia. A cell is a refugium for option k at horizon T only if it is viable in most of the ensemble, topographically decoupled from regional exposure, and inside the area of applicability:

$$R_{ik} = \mathbb{1}[\mathbb{P}_{\omega}(V_{ik} \geq v^*) \geq \rho] \cdot \mathbb{1}[B_i > 0] \cdot \mathbb{1}[\text{DI} \leq \text{DI}^*], \quad B_i = \frac{\bar{E}_{\text{reg},i} - E_i}{\text{sd}(E)}, \quad (8.5)$$

with $v^* = 0.6$, $\rho = 0.8$ as defaults, E a standardised composite of CWD, growing-season VPD and heat exposure, and $\bar{E}_{\text{reg},i}$ its mean within a regional radius. Cells outside the area of applicability are flagged, not asserted. A risk-averse continuous score, $\text{median}(V) - \lambda \text{IQR}(V)$ across the ensemble, ranks cells within the refugium class.

Climate analogues. For each cell and future window we find the present-day cell with the closest climate, using Mahalanobis distance in standardised space (mean and winter-minimum temperature, CWD, VPD, GDD, precipitation seasonality) under the covariance of the reference climate, and the novelty percentile of the distance among reference nearest-neighbour distances (Mahony et al., 2017). Two uses follow. Existing stands at the analogue location are candidate seed sources and provenance tests, with the usual caveat about local adaptation; and no-analogue cells are flagged as places where no statistical or niche model, ours included, should be trusted, which complements the area of applicability with a purely climatic criterion.

9 The machine-learning pipeline and how it is validated

The v1 pipeline failed at evaluation, not at fitting. A random split of spatially autocorrelated pixels rewards a model for memorising place, and the score it produced described a task (interpolating inside the same landscape and era) that nobody needs done. Version 2 follows one rule: a headline number is computed under the resampling design that mimics the use it will be put to, against baselines that could embarrass the model, and every fitted quantity, meaning hyper-parameters, feature choices, stacking weights, the applicability threshold and the recalibration constants of Module B, is fitted inside the training fold only.

Table 4. Validation designs. BSS is the Brier skill score against the stated baseline; the criteria are pre-specified defaults.

Design	What is held out	Question it answers	Pre-specified criterion
Spatial-block CV (5 folds, repeated)	Blocks with side $\geq$ residual-variogram range (25 km until estimated), 2 km buffer, folds balanced by event count	New places, same period	BSS vs. coordinates-only ≥ 0.10 ; calibration slope 0.8–1.2
Forward chaining	All pixels in years $t+1..t+3$ after training on years $\leq t$ (at least 10 training years)	New years	BSS vs. persistence > 0 at every origin
Leave-province-out	One administrative province at a time	Transfer to a management region	BSS > 0 in the worst province
Leave-extreme-year-out	The k driest or hottest years by national SPEI-12	The years that matter most	Observed event count inside the 90 % bootstrap band of predicted
Extrapolation CV	Hottest and driest 30 % of cells on a CWD- $T_{\max}$ composite; train on the rest	Warmer, drier than seen	Gated hybrid log-loss $\leq$ best single component; $\rho_s(\text{DI}, \text{error}) \geq 0.4$
Design-based sample	Independent probability sample of plots (§9.5)	Is the map right, with an honest interval?	Predicted vs. observed per class within ± 0.10

9.1 Three kinds of transfer, six designs

The framework is used to predict in places it has not seen, in years it has not seen and in climates it has not seen. These are different tasks and no single split covers them; Table 4 lists the designs, and the pass criteria are written down now, before any model is fitted. They are defaults to be confirmed at the first gate (§11), not thresholds tuned to a result.

Blocks. The block side is the practical range of the empirical variogram of residuals from a climate-only model, the distance beyond which residuals are effectively independent (Roberts et al., 2017; Valavi et al., 2019). It is then checked with nearest-neighbour distance matching (Milà et al., 2022), which compares the distribution of distances from test to training points in the cross-validation with that from prediction locations to training locations; if the cross-validation is easier than the real task, the blocks are enlarged. Ploton and colleagues showed how badly a random split can overstate large-scale ecological mapping skill (Ploton et al., 2020), and v1’s coordinates-only benchmark (Table 1) is the same lesson at small scale. Meyer and Pebesma (2022) add that a map’s accuracy statement should travel with the domain in which it was validated, which in this framework is the role of the area of applicability.

Nesting. All tuning runs in an inner loop confined to the outer training blocks. The applicability threshold DI^* , the stacking weights and the link-scale recalibration (a_g, b_g) of Module B are treated as model parameters and are refitted in every outer fold, so that no outer test cell has ever influenced them.

9.2 Baselines and metrics

Every design is scored against four baselines: a climatological null (each year’s base rate), persistence (the pixel’s own event history), a coordinates-only model (a smooth spatial function of position and nothing else) and a replica of the v1 random forest. The coordinates-only benchmark matters most. In the v1 data it explained more variance than the climate deltas did (Table 1); a model that cannot beat a map of its own coordinates has learnt *where*, not *why*.

The hazard model is probabilistic, so it is scored with proper scoring rules. For N person-periods with outcomes $y_j \in \{0, 1\}$ and predicted hazards $\hat{p}_j$, weighted by $1/\pi_0$ for sampled non-events,

$$\text{BS} = \frac{1}{N} \sum_j (\hat{p}_j - y_j)^2, \quad \text{BSS} = 1 - \frac{\text{BS}}{\text{BS}_{\text{ref}}}, \quad \text{BS} = \underbrace{\frac{1}{N} \sum_b n_b (\bar{p}_b - \bar{o}_b)^2}_{\text{reliability}} - \underbrace{\frac{1}{N} \sum_b n_b (\bar{o}_b - \bar{o})^2}_{\text{resolution}} + \underbrace{\bar{o}(1 - \bar{o})}_{\text{uncertainty}}, \quad (9.1)$$

with the second identity the Murphy decomposition over probability bins b . Reliability is the part that matters for decisions, because a hazard that is systematically too high or low changes the ranking of options once it is compounded over 30 years. Calibration is also tested directly by regressing the outcome on the logit of the prediction, $\text{logit } \mathbb{P}(y = 1 \mid \hat{p}) = a + b \text{ logit } \hat{p}$, for which the target is $a = 0$, $b = 1$; a slope below 1 signals over-fitting, above 1 under-confidence (Steyerberg et al., 2010). The log-loss is the discrete-time likelihood and doubles as the fitting criterion. Because dieback is rare, precision–recall AUC is reported; ROC AUC is reported but never as a headline, since it ignores both calibration and base rate. Expected calibration error is a summary with a known dependence on binning and is never used alone; cumulative-hazard calibration (predicted against observed cumulative incidence by risk decile) is the check that matches the 30-year use. Overall accuracy and

Cohen's κ are not used (Pontius and Millones, 2011). Continuous outputs (attainable height) are scored by the pinball loss of Eq. (8.1) and by empirical coverage.

9.3 Prediction intervals, and where they stop being valid

For attainable height we wrap the quantile models in conformalised quantile regression (Romano et al., 2019): with lower and upper quantile fits $\hat{q}_{lo}$, $\hat{q}_{hi}$ and calibration scores $E_i = \max\{\hat{q}_{lo}(x_i) - y_i, y_i - \hat{q}_{hi}(x_i)\}$, the interval $[\hat{q}_{lo}(x) - \hat{Q}, \hat{q}_{hi}(x) + \hat{Q}]$ with $\hat{Q}$ the $\lceil (1 - \alpha)(n + 1) \rceil / n$ empirical quantile of E_i has marginal coverage $\geq 1 - \alpha$ under exchangeability (Angelopoulos and Bates, 2023). Under covariate shift the scores can be re-weighted by the density ratio between target and calibration covariates (Tibshirani et al., 2019), but the ratio is undefined where the calibration data have no mass. That is precisely the extrapolation regime, so we do not claim coverage there; intervals outside the area of applicability are drawn hatched, and their empirical coverage is measured in the extrapolation CV and reported.

Applicability is validated as well as used. If the dissimilarity index is informative, prediction error should grow with it; we report skill stratified by DI/DI* in the extrapolation CV, and the gate of Eq. (7.4) is admissible only if the rank correlation between DI and absolute error is at least 0.4. The decay constant $\kappa = 2$ is a default checked against $\kappa \in \{1, 2, 4\}$ on the same held-out tail.

9.4 Why in-range validation cannot rank extrapolators: a controlled experiment

Everything above measures skill on data that resemble the training data. The limits of that are easiest to see in a world where the truth is known. We built two synthetic worlds from the engine of Module B in which mortality hazard depends on hot-spell temperature and a soil-water index. Training data cover $T_{\max} = 24\text{--}34^\circ\text{C}$ only. The worlds differ in a single trait, the temperature T_p at which residual conductance steepens (38°C in world A, 47°C in B), so the data inside the training range are practically indistinguishable, while the hazard beyond it is not (Fig. 4a,b). Four learners are fitted to the same data: a random forest, the cloglog GLM, the mechanistic model alone (deliberately mis-specified by $+2^\circ\text{C}$ in T_p , with wider trait spreads and 15 % higher capacitance) and the AOA-gated hybrid.

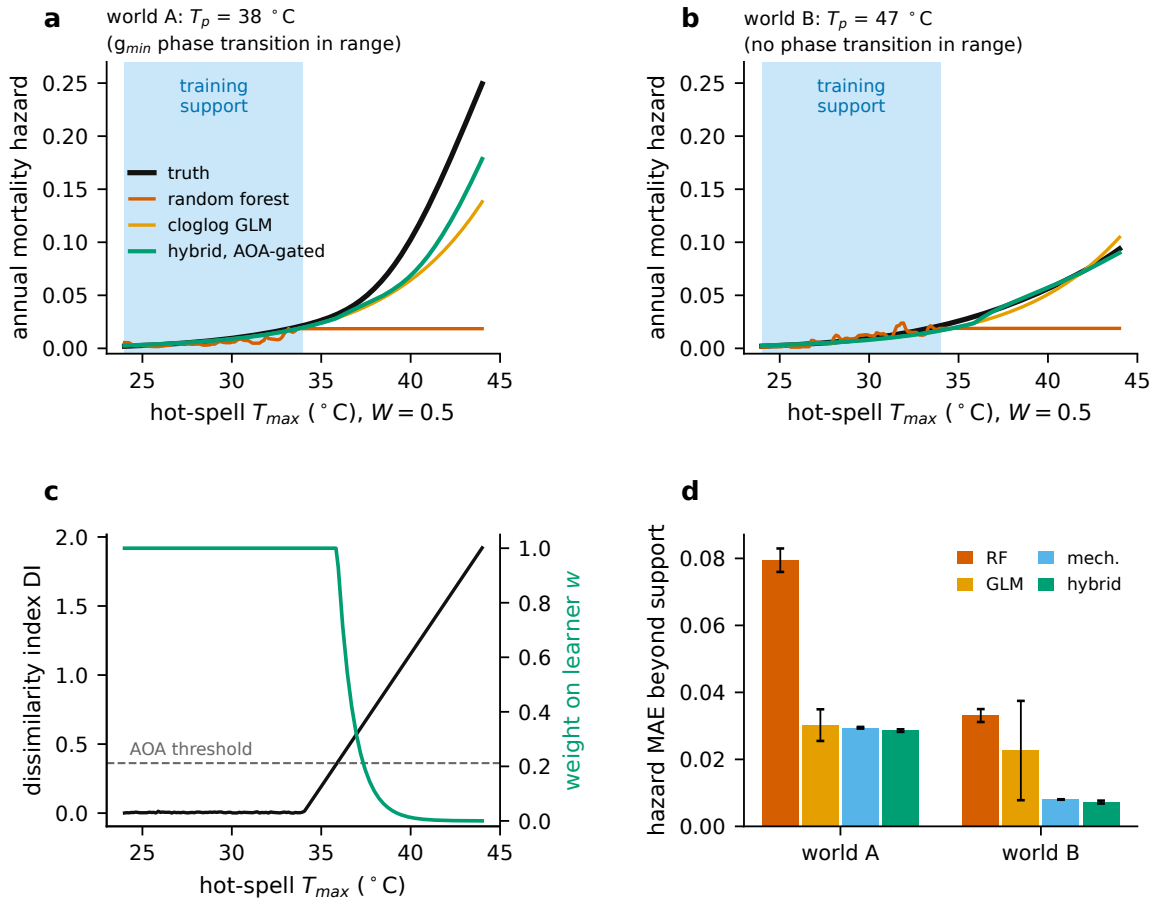
Inside the training range all four methods err by 0.001–0.003 in absolute hazard and cannot be ranked. Beyond it the mean absolute error is, for world A (true mean hazard 0.101), 0.080 for the random forest, 0.030 for the GLM, 0.029 for the mechanistic model and 0.029 for the hybrid; for world B (true mean 0.054), 0.033, 0.023, 0.008 and 0.007. The random forest is worst in both because it holds its last in-range value, as v1's did. The GLM's linear extrapolation on the link scale is an assumption: in B it follows the truth in the typical replicate but its error varies widely (standard deviation 0.015 across replicates), and in A it misses the phase transition. The mechanistic and hybrid models have the lowest error in both worlds, in A only marginally below the GLM, and the hybrid cures nothing: in world A it still under-predicts because its prior places T_p too high, so the transition arrives late. What the design achieves is narrower, and we think more defensible. Extrapolation is made by a model whose form is physically motivated rather than by an accident of the learner; the switch to it is governed by a documented, testable criterion (Fig. 4c); and its remaining error is bounded by the error of the mechanistic prior, which Module B's calibration and sensitivity experiments are built to expose.

9.5 Independent, design-based validation of the products

Cross-validation estimates how a modelling procedure performs on new data; it does not estimate the accuracy of the final map, and with clustered or purposive reference data it can mislead in either direction (Wadoux et al., 2021). The claims that will be made to foresters and funders are claims about maps, so they are backed by an independent probability sample whose inference needs no model of spatial structure (Olofsson et al., 2014; Stehman and Foody, 2019). Strata h are cross-classifications of predicted- V class, area-of-applicability status and elevation zone, with stratum weights W_h from mapped areas. The stratified estimator of a proportion (a dieback frequency, or a survival-and-height frequency) and its variance are

$$\hat{p} = \sum_h W_h \bar{y}_h, \quad \widehat{\text{Var}}(\hat{p}) = \sum_h W_h^2 \frac{s_h^2}{n_h}, \quad n \approx \left(\frac{\sum_h W_h S_h}{S(\hat{p})} \right)^2, \quad S_h = \sqrt{p_h(1 - p_h)}, \quad (9.2)$$

where $S(\hat{p})$ is the target standard error and the last expression is the sample size for that target given anticipated stratum proportions p_h . For four strata with $W = (0.4, 0.3, 0.2, 0.1)$ and $p = (0.05, 0.20, 0.50, 0.80)$, $\sum_h W_h S_h = 0.347$, and a standard error of 0.02 requires $n \approx 300$ plots, against about 480 under simple random sampling; this is the origin of the 300-plot minimum in §4. Allocation is proportional to $W_h S_h$ with a floor of 25 plots per stratum and deliberate over-sampling of strata outside the area of applicability. The plots are fixed before any model is fitted, are excluded from training together with a buffer, and accuracy is reported separately inside and outside the area of applicability.



synthetic data with known truth (10 replicates); mechanistic prior carries a deliberate $+2^\circ\text{C}$ T_p error

Figure 4. Controlled extrapolation experiment on synthetic data with known truth (10 replicates). (a,b) True hazard and model predictions at fixed soil-water index; the shaded band is the training support. (c) Dissimilarity index and gate weight w along the temperature axis; the gate hands over to the mechanistic model as the index passes its threshold. (d) Mean absolute hazard error beyond the training support (mean $\pm$ s.d. of replicates). Illustrative: it demonstrates a mechanism, not the size of the effect in Armenia.

There is one thing no design can do: validate a projection against observations that do not yet exist. What is validated is the machinery that produces projections, in hindcast, in transfer and in extrapolation, and every projected map is accordingly published with its extrapolation footprint instead of an accuracy figure it cannot have.

10 Scenarios, uncertainty and the decision layer

10.1 Scenario and ensemble design

Scenarios are choices about the world, not draws from a distribution, and we do not average them. Results are reported per pathway (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5), with SSP5-8.5 used as a high-end stress test rather than a forecast (Hausfather and Peters, 2020), at 20-year windows centred on 2050, 2080 and 2100 against the fixed 1991–2020 reference. The primary ensemble is the five ISIMIP3b models; NEX-GDDP-CMIP6 supplies SSP2-4.5 and a wider model set to test whether conclusions depend on that particular five.

Two safeguards address the fact that CMIP6 contains models whose sensitivity exceeds the assessed likely range. Every headline result is indexed by global warming level (1.5, 2, 3 and 4 K above pre-industrial) as well as by calendar year, which removes model sensitivity from the argument about *when*; and ensemble statistics are shown with and without any model above the assessed very-likely upper bound of equilibrium sensitivity (Hausfather et al., 2022; Zelinka et al., 2020). Performance-independence weighting (Brunner et al., 2020) is a sensitivity analysis, never the default. Where overshoot runs exist (SSP5-3.4-OS), they serve as a stress test of the lag structure of Module C: if a cohort's viability at 2100 differs between a warming-then-cooling path and a monotone path that ends at the same climate, the framework is representing legacy and hysteresis, the trajectory dependence reported for Amazon dieback under overshoot (Munday et al., 2025).

Table 5. Factors of the uncertainty ensemble.

Factor	Levels	Kind	Treatment
Pathway (SSP)	4, plus overshoot stress test	Deep (a choice)	Conditional reporting; never averaged
Climate model	5 (ISIMIP3b); wider set (NEX-GDDP)	Structural	GWL indexing; with/without hot models
Internal variability	Years within 20-yr windows, realisations	Aleatory	Distribution of annual hazards
Downscaling	QDM, delta change	Structural	Both carried
PET formulation	4 (§5)	Structural	All carried
Soil and rooting depth	50 draws	Epistemic	Random draws from SoilGrids quantiles and priors
Traits	50 hyper-parameter × 200 individual draws	Epistemic and aleatory	Two-level Monte Carlo (§6)
Model class	Statistical, mechanistic, gated hybrid	Structural	All reported; hybrid is the default product

10.2 Partitioning uncertainty

The output V is itself a probability, so the ensemble yields a distribution over probabilities. Its spread has different origins that call for different responses (Table 5), and the framework reports them separately: a scientist can reduce knowledge uncertainty by measuring, a manager can reduce exposure to scenario uncertainty by diversifying, and neither can do anything about internal variability except plan for it.

Analysis of variance for discrete factors. For an output Y (for example regional mean V or the area with $V \geq v^*$) evaluated on the crossed design of the discrete factors, the total variance is split by a standard ANOVA into main effects and interactions, $\text{Var}(Y) = \sum_f \text{Var}_f + \text{interactions}$, and the fractions $F_f = \text{Var}_f / \text{Var}(Y)$ are tracked as a function of lead time in the manner of Hawkins and Sutton (2009) and Lehner et al. (2020). This tells us, for each horizon and region, whether the argument is about emissions, about the climate model, or about our own modelling.

Global sensitivity for the knowledge factors. For the continuous, epistemic factors $X = (X_1, \dots, X_k)$ (rooting depth, plant-available water, trait hyper-parameters, recalibration constants) we use variance-based indices computed on the emulator of Module B, since the full engine is too slow for the number of evaluations needed. The first-order and total-order indices are

$$S_i = \frac{\text{Var}[\mathbb{E}(Y | X_i)]}{\text{Var}(Y)}, \quad S_{T_i} = \frac{\mathbb{E}[\text{Var}(Y | X_{\sim i})]}{\text{Var}(Y)} \approx \frac{1}{2N \widehat{\text{Var}}(Y)} \sum_{j=1}^N [f(A)_j - f(A_B^{(i)})_j]^2, \quad (10.1)$$

with the Jansen-type total-order estimator recommended by Saltelli et al. (2010), in which A and B are independent sample matrices and $A_B^{(i)}$ is A with column i taken from B ; the cost is $N(k+2)$ emulator runs. A large $S_{T_i} - S_i$ marks interactions, which in a threshold system like this one (phase transition, stomatal closure) are expected and should not be hidden. The result is a ranked list of which measurement would reduce uncertainty most, and it sets the priorities of the trait campaign in §11.

10.3 Decision layer: robust portfolios

Ranking species by expected viability cell by cell yields a landscape planted to whichever option is best on average, and that landscape fails all at once in the scenarios where the option is not. A planting programme is a portfolio, so it is optimised as one.

Let u index management units (aggregated 30 m cells, roughly 1 ha), j planting options, and $\omega \in \Omega$ the members of the scenario ensemble. Let $x_{uj} \in \{0, 1\}$ indicate that option j is planted in unit u , A_u its area, and $b_{uj\omega} = V_{uj\omega} v_j$ the benefit per hectare, viability times a value weight v_j (canopy area, carbon or water services, set with stakeholders). The benefit of a plan in scenario ω is $B_\omega(x) = \sum_{u,j} A_u b_{uj\omega} x_{uj}$. The plan maximises a mixture of the mean and the mean of the worst tail,

$$\max_{x, \zeta, s} (1 - \lambda) \frac{1}{|\Omega|} \sum_{\omega} B_\omega(x) + \lambda \left[\zeta - \frac{1}{(1 - \alpha)|\Omega|} \sum_{\omega} s_\omega \right] \quad \text{s.t.} \quad s_\omega \geq \zeta - B_\omega(x), \quad s_\omega \geq 0, \quad (10.2)$$

where the bracket is the conditional value-at-risk, the mean benefit over the worst $(1 - \alpha)$ share of scenarios, in the linear form of Rockafellar and Uryasev (2000). The remaining constraints are

$$\sum_j x_{uj} \leq 1, \quad \sum_{u,j} A_u c_{uj} x_{uj} \leq \text{Budget}, \quad \sum_u A_u x_{uj} \leq \sigma_{\max} \sum_{u,j'} A_u x_{uj'}, \quad x_{uj} \leq e_{uj}, \quad (10.3)$$

Table 6. Work packages. Months are from project start.

Work package	Months	Main outputs
WP1 Data, labels, harmonisation	1–8	Master grid and data manifests; versioned dieback panel; plot and ring database
WP2 Module A: topoclimate and water balance	2–8	30 m forcing ensemble, PET ensemble, ψ_s ; station cross-validation and closure tests
WP3 Traits and hydraulic engine	3–14	Trait posteriors; local P_{50} , $g_{\min}(T)$, pressure–volume curves for 4–6 species; emulator; ABC calibration
WP4 Mortality hazard model	6–14	Hazard stack, DI-based area of applicability, nested CV report
WP5 Growth, establishment, plantation survey	4–18	Attainable-height and growth models; establishment hazard; two field seasons of planting-follow-up plots
WP6 Uncertainty, viability and decision	12–22	Ensemble V , refugia, analogues; ANOVA and Sobol’ partition; portfolio tool
WP7 Validation, products and dissemination	1–24	Design-based sample (months 14–20); model cards; papers; training for Armenian partners

one option per unit, a budget, a cap $\sigma_{\max}$ on the area share of any single option (response diversity), and an eligibility indicator e_{uj} that is zero where the option is outside its applicability region or legally excluded. A per-basin cap on additional water use can be added as a further linear constraint. The problem is a mixed-integer linear programme, solved with HiGHS in the skeleton. With about 500 scenario members, $\alpha = 0.8$ (the worst 100), a few thousand units and ten to fifteen options, it has of the order of 10^4 – 10^5 binary variables, which is tractable after aggregation and, if necessary, scenario reduction.

Sweeping λ from 0 to 1 traces the mean–CVaR frontier, and the distance between its ends is the price of robustness, expressed in units of benefit (hectares of expected surviving canopy for canopy-area weights). The optimiser exploits whatever its scenario sample happens to contain, so the reported CVaR is evaluated on a fresh, independent set of ensemble members; a large gap between the in-sample and out-of-sample values is a warning that the portfolio is over-fitted to the ensemble. Areas with a large extrapolation footprint are not silently planted or silently excluded: they are flagged, and the plan states how much of its expected benefit comes from cells outside the area of applicability.

11 Work plan, gates, outputs and what would prove us wrong

11.1 Work packages and go/no-go gates

The plan is indicative and expressed in months from project start (24 in total); the calendar follows the funding decision. It is ordered so that the cheapest and most decisive questions are answered first: whether the physical drivers are right (WP2), and whether observed mortality can be predicted better than by a map of coordinates (WP4). The expensive parts, the field survey and the full ensemble, start only after the gates that justify them.

Four gates decide whether the work continues as planned, and each names its fallback so that a failure is a result rather than a crisis.

G1, month 8: physics and data are adequate. Module A closes its water balance, its downscaled temperature and precipitation reduce the raw ERA5-Land error at held-out stations, and it reproduces TerraClimate CWD within a stated tolerance; the minimum data of §4 exist. *Fallback:* coarsen to 100 m and extend the label network before fitting anything.

G2, month 14: the hazard model earns its place. Under spatial-block CV and forward chaining it meets the criteria of Table 4 against the coordinates-only and persistence baselines. *Fallback:* the statistical component is retired to the role of benchmark, the mechanistic model is reported with widened uncertainty, and the failure itself, a negative result on dieback predictability from climate in this region, is published.

G3, month 18: the gate is justified. In the extrapolation CV the hybrid is no worse than the better of its components, $\rho_s(\text{DI}, \text{error}) \geq 0.4$, and the emulator reproduces the full engine to within 0.02 in hazard. *Fallback:* report mechanistic and statistical bounds separately instead of a blended hazard.

G4, month 20: products are validated. The design-based sample is complete and predicted V agrees with observed class frequencies within ± 0.10 , inside and outside the area of applicability separately. Maps and portfolios are released only after G4; before that they are labelled as provisional.

11.2 Deliverables and publication plan

The project produces an open-source package (the skeleton of §11.4 developed to full implementation), versioned datasets of dieback labels and trait posteriors, viability, refugium and extrapolation-footprint layers with the uncertainty partition, a portfolio tool, a model card per product stating intended use, training data, validation

Table 7. Risk register (main items).

Risk	Consequence	Mitigation and fallback
Satellite dieback labels are too noisy	Biased hazards, weak signal	Misclassification correction (Eq. 7.1); tree-ring collapses as second response; interpret-and-verify sample
No local hydraulic traits for Caucasian species	Module B leans on priors	Hierarchical priors from XFT/TRY; measurement campaign for 4–6 species, targeted by the Sobol' ranking
Little independent plantation data	Establishment hazard weakly identified	Prioritise WP5 survey; use plot regeneration data; report V without the establishment factor where unsupported
Compute at 30 m × ensemble	Schedule slips	Emulator; full ensemble at 100 m; 30 m only for median and decision subsets
Station data hard to obtain	Weaker Module A validation	ERA5-Land reference with explicit sensitivity; validate on TerraClimate and satellite soil moisture
Hybrid does not beat the statistical model	Gate is unnecessary	Report as a negative result; the gate reduces to a flag on the extrapolation footprint

results and known failure modes, and technical reports for the FORACCA partners. Three papers are planned; each stands on its own and none depends on the outcome of the others. Target venues are indicative and depend on the gate results.

1. **A hazard-model paper:** discrete-time cloglog survival modelling of forest dieback with within/between decomposition, drought legacies and label-error correction, for Armenia. Its central claim is methodological and testable: what the space-for-time assumption changes, measured by the width of the bracket in Eq. (7.5). Candidate outlets: *Global Change Biology*, *Ecological Applications*.
2. **A methods paper:** gating a mechanistic and a statistical hazard model by area of applicability, with the controlled experiment of Fig. 4 extended to real held-out climates. Candidate outlet: *Methods in Ecology and Evolution*.
3. **A decision paper:** robust species–site portfolios under partitioned uncertainty, with the price of robustness and the value of measuring traits from the Sobol' analysis. Candidate outlets: *Nature Sustainability*, *Nature Communications*, or a specialist journal if the results warrant less.

11.3 Principal risks

The six risks that most threaten the schedule or the claims are listed in Table 7 with their fallbacks; each is tied to a gate above, so that a problem surfaces at a decision point rather than in the final year.

11.4 The repository skeleton

Alongside this document we provide a Python package, `foracca2`, that fixes the interfaces and implements the numerical kernels that the rest depends on: vapour-pressure and VPD algebra, slope radiation, a snow and soil bucket with mass-balance closure, PET formulations, the vulnerability curve, two-phase engine, $g_{\min}(T)$ and Monte Carlo failure probability, the cloglog hazard model with DLNM lags, the applicability index and gate, block, forward-chaining and group splits with variogram range, proper scoring rules, conformal intervals, design-based estimators, ANOVA and Sobol' analysis, the cohort viability and refugium logic, and the CVaR portfolio optimiser. About 1,700 lines of package code are covered by 77 tests, including closed-form checks (the two-phase engine against $B/E_{\min}$, the Ishigami function for Sobol' indices, mass balance to machine precision), physical bounds on derived quantities and regression tests that encode the v1 errors. Configuration files hold the scenario, validation and study-area choices; a Snakemake workflow declares the contract between modules. Data access (Earth Engine export) and the full-scale pipeline steps are declared but not implemented, and the species trait file contains clearly labelled placeholders that exercise the code and are not results.

11.5 Limitations, and what would falsify the framework

The framework has limits that are written down here so that they are not discovered later. Biotic mortality is a climate-independent baseline and is not projected. Carbon starvation, root–shoot dynamics and the CO_2 effect on water-use efficiency enter only through scenario knobs. Local adaptation and provenance effects are represented by trait classes rather than by genetics. Satellite dieback is a proxy for mortality whose error we model but cannot remove, and the number of Armenian tree-ring chronologies is small. The scenario ensemble is a set of stories rather than a probability distribution, and future extremes outside any observed range rest on physics that we can test in hindcast and in synthetic worlds but not in the future.

Because the specification names its own tests, it can also fail visibly. It would be wrong, or at least not worth its complexity, if the coordinates-only baseline matched the hazard model under forward chaining; if calibration slopes lay outside 0.7–1.3 after recalibration; if the Boyce index of predicted viability against adult occurrence

were not positive; if the dissimilarity index were unrelated to error in the extrapolation CV; or if the design-based sample showed that V is systematically higher than the observed survival-and-height frequency. Any of these would be reported, and the corresponding module would revert to its documented fallback.

References

- John T. Abatzoglou, Solomon Z. Dobrowski, Sean A. Parks, and Katherine C. Hegewisch. TerraClimate, a high-resolution global dataset of monthly climate and climatic water balance from 1958–2015. *Scientific Data*, 5:170191, 2018.
- Henry D. Adams et al. A multi-species synthesis of physiological mechanisms in drought-induced tree mortality. *Nature Ecology & Evolution*, 1:1285–1291, 2017.
- Craig D. Allen et al. A global overview of drought and heat-induced tree mortality reveals emerging climate change risks for forests. *Forest Ecology and Management*, 259:660–684, 2010.
- Richard G. Allen, Luis S. Pereira, Dirk Raes, and Martin Smith. Crop evapotranspiration: guidelines for computing crop water requirements. Technical Report 56, FAO, 1998.
- William R. L. Anderegg et al. Pervasive drought legacies in forest ecosystems and their implications for carbon cycle models. *Science*, 349:528–532, 2015.
- Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning*, 16:494–591, 2023.
- Mark A. Beaumont. Approximate bayesian computation in evolution and ecology. *Annual Review of Ecology, Evolution, and Systematics*, 41:379–406, 2010.
- Santiago Beguéria, Sergio M. Vicente-Serrano, Fergus Reig, and Borja Latorre. Standardized precipitation evapotranspiration index (SPEI) revisited: parameter fitting, evapotranspiration models, tools, datasets and drought monitoring. *International Journal of Climatology*, 34:3001–3023, 2014.
- Andrew Bell and Kelvyn Jones. Explaining fixed effects: random effects modeling of time-series cross-sectional and panel data. *Political Science Research and Methods*, 3:133–153, 2015.
- Amy C. Bennett, Nate G. McDowell, Craig D. Allen, and Kristina J. Anderson-Teixeira. Larger trees suffer most during drought in forests worldwide. *Nature Plants*, 1:15139, 2015.
- Leo Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996.
- Tim J. Brodribb and Hervé Cochard. Hydraulic failure defines the recovery and point of death in water-stressed conifers. *Plant Physiology*, 149:575–584, 2009.
- Lukas Brunner et al. Reduced global warming from CMIP6 projections when weighting models by performance and independence. *Earth System Dynamics*, 11:995–1012, 2020.
- Brian S. Cade and Barry R. Noon. A gentle introduction to quantile regression for ecologists. *Frontiers in Ecology and the Environment*, 1:412–420, 2003.
- Gustau Camps-Valls et al. A unified vegetation index for quantifying the terrestrial biosphere. *Science Advances*, 7:eabc7447, 2021.
- Alex J. Cannon, Stephen R. Sobie, and Trevor Q. Murdock. Bias correction of GCM precipitation by quantile mapping: how well do methods preserve changes in quantiles and extremes? *Journal of Climate*, 28:6938–6959, 2015.
- Brendan Choat et al. Global convergence in the vulnerability of forests to drought. *Nature*, 491:752–755, 2012.
- Brendan Choat et al. Triggers of tree mortality under drought. *Nature*, 558:531–539, 2018.
- Roger B. Clapp and George M. Hornberger. Empirical equations for some soil hydraulic properties. *Water Resources Research*, 14:601–604, 1978.
- Martin Claverie et al. The Harmonized Landsat and Sentinel-2 surface reflectance data set. *Remote Sensing of Environment*, 219:145–161, 2018.
- Hervé Cochard. A new mechanism for tree mortality due to drought and heatwaves. bioRxiv preprint 531632, 2019.
- Hervé Cochard, François Pimont, Julien Ruffault, and Nicolas Martin-StPaul. SurEau: a mechanistic model of plant water relations under extreme drought. *Annals of Forest Science*, 78:55, 2021.
- Dağtekin, Şahan, Denk, Köse, and Dalfes. Past, present and future distributions of Oriental beech (*fagus orientalis*) under climate change projections. *PLOS ONE*, 15(11):e0242280, 2020.
- Kimberley T. Davis et al. Wildfires and climate change push low-elevation forests across a critical climate threshold for tree regeneration. *Proceedings of the National Academy of Sciences*, 116:6193–6198, 2019.
- Solomon Z. Dobrowski et al. Forest structure and species traits mediate projected recruitment declines in western US tree species. *Global Ecology and Biogeography*, 24:917–927, 2015.
- Jeff Dozier and James Frew. Rapid calculation of terrain parameters for radiation modeling from digital elevation data. *IEEE Transactions on Geoscience and Remote Sensing*, 28:963–969, 1990.
- Ralph Dubayah et al. The Global Ecosystem Dynamics Investigation: high-resolution laser ranging of the Earth’s forests and topography. *Science of Remote Sensing*, 1:100002, 2020.
- Remko A. Duursma et al. On the minimum leaf conductance: its role in models of plant water use, and ecological and environmental controls. *New Phytologist*, 221:693–705, 2019.
- Jane Elith, Michael Kearney, and Steven Phillips. The art of modelling range-shifting species. *Methods in Ecology and*

- Evolution*, 1:330–342, 2010.
- Matthew C. Fitzpatrick and William W. Hargrove. The projection of species distribution models and the problem of non-analog climate. *Biodiversity and Conservation*, 18:2255–2261, 2009.
- Antonio Gasparrini, Ben Armstrong, and Michael G. Kenward. Distributed lag non-linear models. *Statistics in Medicine*, 29: 2224–2234, 2010.
- Alice Gauthey. Beyond P50, a better way to predict tree mortality. *Plant Physiology*, 201:kiag204, 2026.
- André Granier, Nathalie Bréda, Pierre Biron, and Sylvie Villette. A lumped water balance model to evaluate duration and intensity of drought constraints in forest stands. *Ecological Modelling*, 116:269–283, 1999.
- Volker Grimm et al. Pattern-oriented modeling of agent-based complex systems: lessons from ecology. *Science*, 310:987–991, 2005.
- Charlotte Grossiord. Having the right neighbors: how tree species diversity modulates drought impacts on forests. *New Phytologist*, 228:42–49, 2020.
- Antoine Guisan and Wilfried Thuiller. Predicting species distribution: offering more than simple habitat models. *Ecology Letters*, 8:993–1009, 2005.
- Matthew C. Hansen et al. High-resolution global maps of 21st-century forest cover change. *Science*, 342:850–853, 2013.
- Henrik Hartmann et al. Research frontiers in drought-induced tree mortality: crossing scales and disciplines. *New Phytologist*, 218:15–28, 2018.
- Zeke Hausfather and Glen P. Peters. Emissions – the ‘business as usual’ story is misleading. *Nature*, 577:618–620, 2020.
- Zeke Hausfather, Kate Marvel, Gavin A. Schmidt, John W. Nielsen-Gammon, and Mark Zelinka. Climate simulations: recognize the ‘hot model’ problem. *Nature*, 605:26–29, 2022.
- Ed Hawkins and Rowan Sutton. The potential to narrow uncertainty in regional climate predictions. *Bulletin of the American Meteorological Society*, 90:1095–1108, 2009.
- Alexandre H. Hirzel, Gwénaëlle Le Lay, Veronique Helfer, Christophe Randin, and Antoine Guisan. Evaluating the ability of habitat suitability models to predict species presences. *Ecological Modelling*, 199:142–152, 2006.
- Stephen T. Jackson, Julio L. Betancourt, Robert K. Booth, and Stephen T. Gray. Ecology and the ratchet of events: climate variability, niche dimensions, and species distributions. *Proceedings of the National Academy of Sciences*, 106:19685–19692, 2009.
- Dirk N. Karger et al. Climatologies at high resolution for the earth’s land surface areas. *Scientific Data*, 4:170122, 2017.
- Jürgen Kattge et al. TRY plant trait database – enhanced coverage and open access. *Global Change Biology*, 26:119–188, 2020.
- Michael Kearney and Warren Porter. Mechanistic niche modelling: combining physiological and spatial data to predict species’ ranges. *Ecology Letters*, 12:334–350, 2009.
- Robert E. Kennedy, Zhiqiang Yang, and Warren B. Cohen. Detecting trends in forest disturbance and recovery using yearly Landsat time series: 1. LandTrendr – temporal segmentation algorithms. *Remote Sensing of Environment*, 114:2897–2910, 2010.
- Gary King and Langche Zeng. Logistic regression in rare events data. *Political Analysis*, 9:137–163, 2001.
- Joe J. Landsberg and Richard H. Waring. A generalised model of forest productivity using simplified concepts of radiation-use efficiency, carbon balance and partitioning. *Forest Ecology and Management*, 95:209–228, 1997.
- Nico Lang, Walter Jetz, Konrad Schindler, and Jan Dirk Wegner. A high-resolution canopy height model of the Earth. *Nature Ecology & Evolution*, 7:1778–1789, 2023.
- Stefan Lange. Trend-preserving bias adjustment and statistical downscaling with ISIMIP3BASD (v1.0). *Geoscientific Model Development*, 12:3055–3070, 2019.
- Stefan Lange and Matthias Büchner. ISIMIP3b bias-adjusted atmospheric climate input data (v1.1). ISIMIP Repository, 2021.
- Flavio Lehner et al. Partitioning climate projection uncertainty with multiple large ensembles and CMIP5/6. *Earth System Dynamics*, 11:491–508, 2020.
- James A. Lutz, Jan W. van Wageningen, and Jerry F. Franklin. Climatic water deficit, tree species ranges, and climate change in Yosemite National Park. *Journal of Biogeography*, 37:936–950, 2010.
- Colin R. Mahony, Alex J. Cannon, Tongli Wang, and Sally N. Aitken. A closer look at novel climates: new methods and insights at continental to landscape scales. *Global Change Biology*, 23:3934–3955, 2017.
- Nicolas Martin-StPaul, Sébastien Delzon, and Hervé Cochard. Plant resistance to drought depends on timely stomatal closure. *Ecology Letters*, 20:1437–1447, 2017.
- Nate McDowell et al. Mechanisms of plant survival and mortality during drought: why do some plants survive while others succumb to drought? *New Phytologist*, 178:719–739, 2008.
- Hanna Meyer and Edzer Pebesma. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12:1620–1633, 2021.
- Hanna Meyer and Edzer Pebesma. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Communications*, 13:2208, 2022.
- Carles Milà, Jorge Mateu, Edzer Pebesma, and Hanna Meyer. Nearest neighbour distance matching leave-one-out cross-validation for map validation. *Methods in Ecology and Evolution*, 13:1304–1316, 2022.
- Paul C. D. Milly and Krista A. Dunne. Potential evapotranspiration and continental drying. *Nature Climate Change*, 6:946–949, 2016.

- John L. Monteith. Evaporation and environment. *Symposia of the Society for Experimental Biology*, 19:205–234, 1965.
- Munday et al. Risks of unavoidable impacts on forests at 1.5 °c with and without overshoot. *Nature Climate Change*, 15: 650–655, 2025.
- Yair Mundlak. On the pooling of time series and cross section data. *Econometrica*, 46:69–85, 1978.
- Joaquín Muñoz-Sabater et al. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13:4349–4383, 2021.
- Pontus Olofsson, Giles M. Foody, Martin Herold, Stephen V. Stehman, Curtis E. Woodcock, and Michael A. Wulder. Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment*, 148:42–57, 2014.
- Magdalena Opála-Owczarek et al. Dendrochronological potential of drought-sensitive tree stands in Armenia for the hydroclimate reconstruction of the Lesser Caucasus. *Atmosphere*, 12(2):153, 2021.
- Norman W. Pammenter and Christiaan Van der Willigen. A mathematical and statistical analysis of the curves illustrating vulnerability of xylem to cavitation. *Tree Physiology*, 18:589–593, 1998.
- Richard G. Pearson and Terence P. Dawson. Predicting the impacts of climate change on the distribution of species: are bioclimate envelope models useful? *Global Ecology and Biogeography*, 12:361–371, 2003.
- Pierre Ploton et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11:4540, 2020.
- Laura Poggio et al. SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL*, 7: 217–240, 2021.
- Robert G. Pontius and Marco Millones. Death to Kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment. *International Journal of Remote Sensing*, 32:4407–4429, 2011.
- Ross L. Prentice and Larry A. Gloeckler. Regression analysis of grouped survival data with application to breast cancer data. *Biometrics*, 34:57–67, 1978.
- Charles H. B. Priestley and Robert J. Taylor. On the assessment of surface heat flux and evaporation using large-scale parameters. *Monthly Weather Review*, 100:81–92, 1972.
- R. Pshegusov, F. Tembotova, V. Chadaeva, Y. Sablirova, M. Mollaeva, and A. Akhomgotov. Ecological niche modeling of the main forest-forming species in the Caucasus. *Forest Ecosystems*, 9:100019, 2022.
- Markus Reichstein et al. Deep learning and process understanding for data-driven Earth system science. *Nature*, 566:195–204, 2019.
- David R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40:913–929, 2017.
- R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2:21–42, 2000.
- Walter J. Rogan and Beth Gladen. Estimating prevalence from the results of a screening test. *American Journal of Epidemiology*, 107:71–76, 1978.
- Yaniv Romano, Evan Patterson, and Emmanuel Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Julien Ruffault, François Pimont, Hervé Cochard, Jean-Luc Dupuy, and Nicolas Martin-StPaul. SurEau-Ecos v2.0: a trait-based plant hydraulics model for simulations of plant water status and drought-induced mortality at the ecosystem level. *Geoscientific Model Development*, 15:5593–5626, 2022.
- Steven W. Running, Ramakrishna R. Nemani, and Roger D. Hungerford. Extrapolation of synoptic meteorological data in mountainous terrain and its use for simulating forest evapotranspiration and photosynthesis. *Canadian Journal of Forest Research*, 17:472–483, 1987.
- Andrea Saltelli, Paola Annoni, Ivano Azzini, Francesca Campolongo, Marco Ratto, and Stefano Tarantola. Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index. *Computer Physics Communications*, 181:259–270, 2010.
- Keith E. Saxton and Walter J. Rawls. Soil water characteristic estimates by texture and organic matter for hydrologic solutions. *Soil Science Society of America Journal*, 70:1569–1578, 2006.
- Judith D. Singer and John B. Willett. *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence*. Oxford University Press, 2003.
- Jens Peter Skovsgaard and Jerome K. Vanclay. Forest site productivity: a review of the evolution of dendrometric concepts for even-aged stands. *Forestry*, 81:13–31, 2008.
- Julia A. Sohn, Somidh Saha, and Jürgen Bauhus. Potential of forest thinning to mitigate drought stress: a meta-analysis. *Forest Ecology and Management*, 380:261–273, 2016.
- Stephen V. Stehman and Giles M. Foody. Key issues in rigorous accuracy assessment of land cover products. *Remote Sensing of Environment*, 231:111199, 2019.
- Stepanyan et al. Accelerating warming in Armenia (South Caucasus) shifts the climate–growth relationships of *fagus orientalis* L. *Ecologies*, 7(2):32, 2026.
- Nathan L. Stephenson. Actual evapotranspiration and deficit: biologically meaningful correlates of vegetation distribution across spatial scales. *Journal of Biogeography*, 25:855–870, 1998.
- Ewout W. Steyerberg et al. Assessing the performance of prediction models: a framework for traditional and novel measures. *Epidemiology*, 21:128–138, 2010.

- Bridget Thrasher, Weile Wang, Andrew Michaelis, Forrest Melton, Tsengdar Lee, and Ramakrishna Nemani. NASA global daily downscaled projections, CMIP6. *Scientific Data*, 9:262, 2022.
- Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Morgane Urli, Annabel J. Porté, Hervé Cochard, Yann Guengant, Régis Burlett, and Sébastien Delzon. Xylem embolism threshold for catastrophic hydraulic failure in angiosperm trees. *Tree Physiology*, 33:672–683, 2013.
- Roosbeh Valavi, Jane Elith, José J. Lahoz-Monfort, and Gurutzeta Guillera-Arroita. blockCV: an R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, 10:225–232, 2019.
- Sergio M. Vicente-Serrano, Santiago Beguéria, and Juan I. López-Moreno. A multiscalar drought index sensitive to global warming: the standardized precipitation evapotranspiration index. *Journal of Climate*, 23:1696–1718, 2010.
- Voß, Skiadaresis, Linne, Matsakyan, and Seim. Drought limits growth and hydraulic capacity of two ecologically and economically important tree species in the Lesser Caucasus. *Dendrochronologia*, 91, 2025.
- Alexandre M. J.-C. Wadoux, Gerard B. M. Heuvelink, Sytze de Bruin, and Dick J. Brus. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecological Modelling*, 457:109692, 2021.
- Waite et al. Coordinated hydraulic traits predict the time to critical hydraulic failure in temperate broad-leaved and conifer tree species. *Plant Physiology*, 2026. advance article, kiag177.
- David H. Wolpert. Stacked generalization. *Neural Networks*, 5:241–259, 1992.
- Yuting Yang, Michael L. Roderick, Shulei Zhang, Tim R. McVicar, and Randall J. Donohue. Hydrologic implications of vegetation response to elevated CO₂ in climate projections. *Nature Climate Change*, 9:44–48, 2019.
- Mark D. Zelinka et al. Causes of higher climate sensitivity in CMIP6 models. *Geophysical Research Letters*, 47:e2019GL085782, 2020.
- Zhe Zhu and Curtis E. Woodcock. Continuous change detection and classification of land cover using all available Landsat data. *Remote Sensing of Environment*, 144:152–171, 2014.